SpiceMCP circuit catalog

All 25 circuits built so far, in serial order: schematic, simulated response and key result.

No.	Circuit	Type	Key result
1	**RC low-pass filter**	ready-made circuit	Cutoff (-3 dB) 1000.0 Hz for a 1 kHz spec
2	**Sallen-Key low-pass filter**	ready-made circuit	-3 dB at 9998 Hz for a 10 kHz spec, flat passband (peak 0.001 dB)
3	**Non-inverting amplifier**	ready-made circuit	Gain 20.00 dB (x10.00) for a x10 spec
4	**LDO voltage regulator**	ready-made circuit	3.301 V out from 5 V in, at 0.1 A
5	**Buck DC-DC converter (closed loop)**	ready-made circuit	4.999 V out from 12 V in, at 1 A, 500 kHz switching
6	**Instrumentation amplifier**	designed from scratch	Gain 40.00 dB (x100) for a x100 spec
7	**Inverting amplifier**	drawing test	Gain 19.98 dB (x−9.98), set by R2/R1 = 10
8	**Common-source MOSFET amplifier**	drawing test	Mid-band gain -2.0 dB (x0.80) from one MOSFET biased at 0.11 mA
9	**Wheatstone bridge**	drawing test	A 10% change in one arm gives 216 mV across the bridge at 10 V
10	**Boost converter**	drawing test	Steps 5 V up to 9.28 V (ideal at 50% duty: 10 V) into 50 Ω
11	**Two-stage amplifier with feedback**	drawing test	Gain 20.00 dB (x10.00), set by 1 + R2/R3 = 10
12	**Pi filter**	drawing test	Passband -6.1 dB (the 50 Ω source and load halve the voltage)
13	**5-stage RC ladder**	drawing test	-3 dB at 13.2 kHz (each section loads the one before it)
14	**LC tank**	drawing test	Resonates at 5.01 MHz (1/2π√LC = 5.03 MHz)
15	**Two-stage CMOS op-amp**	building block	Open-loop gain 87.5 dB
16	**Comparator with gate drive**	building block	Output swings rail to rail (0 to 12 V) whenever in+ crosses in−
17	**Chua's chaotic circuit**	demo	Chaotic oscillator: L, two capacitors and a negative-resistance (Chua) diode
18	**Inverting amplifier (template)**	ready-made circuit	Gain 20.00 dB (x−9.995) for a x−10 spec, input resistance 10 kΩ
19	**Band-pass filter**	ready-made circuit	Centre gain -0.01 dB at 1 kHz for a 0 dB spec (peak -0.01 dB)
20	**Current-sense amplifier**	ready-made circuit	2.487 V out at 2 A for a 2.5 V full scale (error -0.53%)
21	**ADC driver**	ready-made circuit	Settles to 0.06 LSB of its final value within the acquisition time after a 2 V step (spec: half an LSB)
22	**Bandgap voltage reference**	ready-made circuit	2.4995 V at 25 °C for a 2.5 V spec
23	**Input protection**	ready-made circuit	Clamps a 40 V surge to 16.7 V (load max 20 V)

No.	Circuit	Type	Key result
24	**Boost converter (closed loop)**	ready-made circuit	11.997 V out from 5 V in, at 0.5 A, 300 kHz, closed loop
25	**Flyback converter (closed loop)**	ready-made circuit	4.997 V out from 12 V in, at 1 A, 200 kHz, closed loop

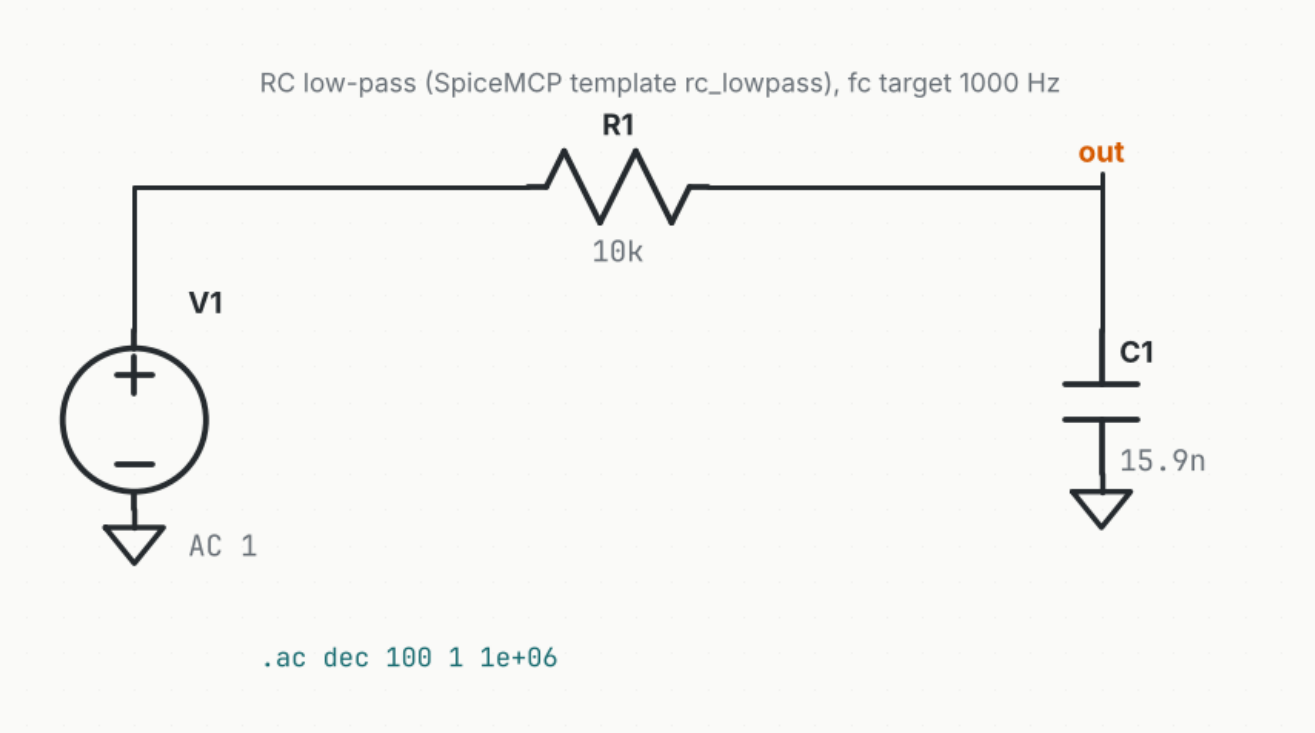
How to read the pages

- **Ready-made circuits (1–5, 18–25)**: you give a spec, SpiceMCP sizes the parts.
- **Designed from scratch (6)**: built with the general tools, not the library.
- **Drawing tests (7–14)**: circuits the schematic drawer was never tuned on.
- **Building blocks (15–16)**: the transistor-level parts inside the others.
- Schematics 1–14 were drawn by SpiceMCP from the simulated netlist and read back by LTspice 26 as the same circuit; 18–24 are drawn the same way (not yet re-read in LTspice). The flyback (25) is not drawn: the auto-placer has no transformer symbol. Plots are ngspice simulations of that netlist.

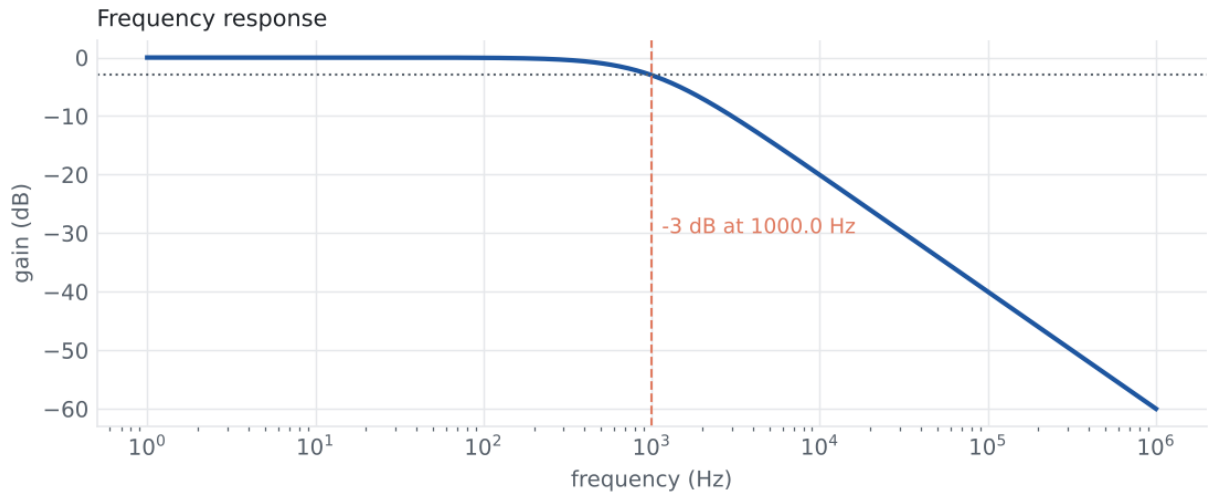
CIRCUIT 1 OF 25 · READY-MADE CIRCUIT

1. RC low-pass filter

First-order passive filter: lets low frequencies through, cuts high ones at 20 dB per decade.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

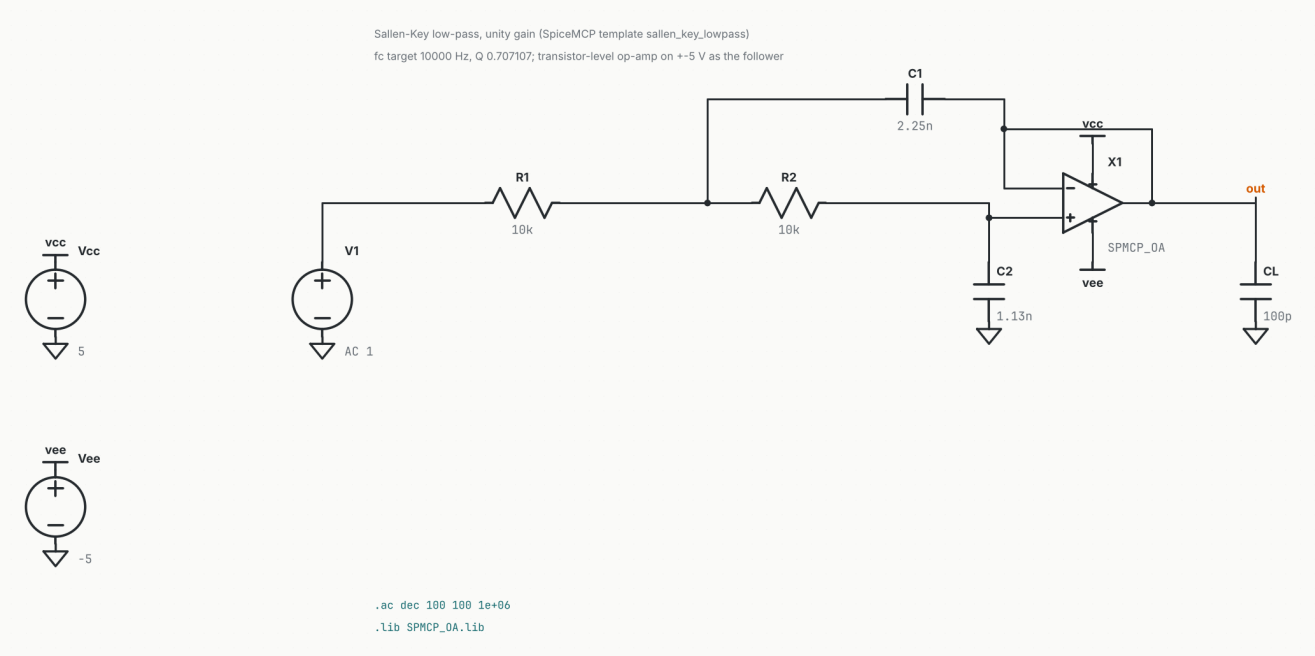
KEY RESULT

- Cutoff (-3 dB) 1000.0 Hz for a 1 kHz spec
- -20.0 dB one decade above cutoff (20 dB/decade)

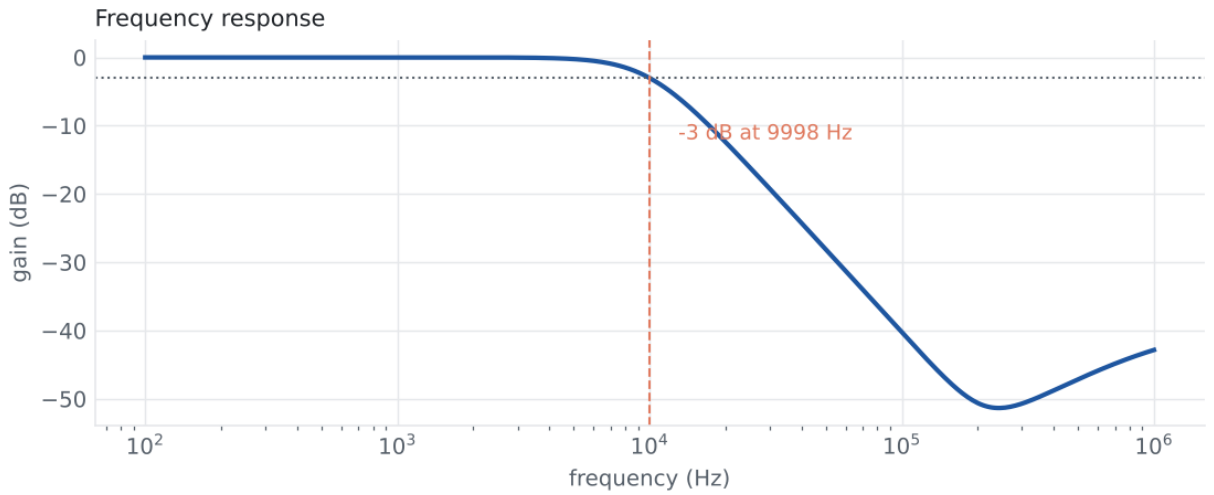
CIRCUIT 2 OF 25 · READY-MADE CIRCUIT

2. Sallen-Key low-pass filter

Second-order active filter (Butterworth, Q = 0.707) around the transistor-level op-amp: 40 dB per decade.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

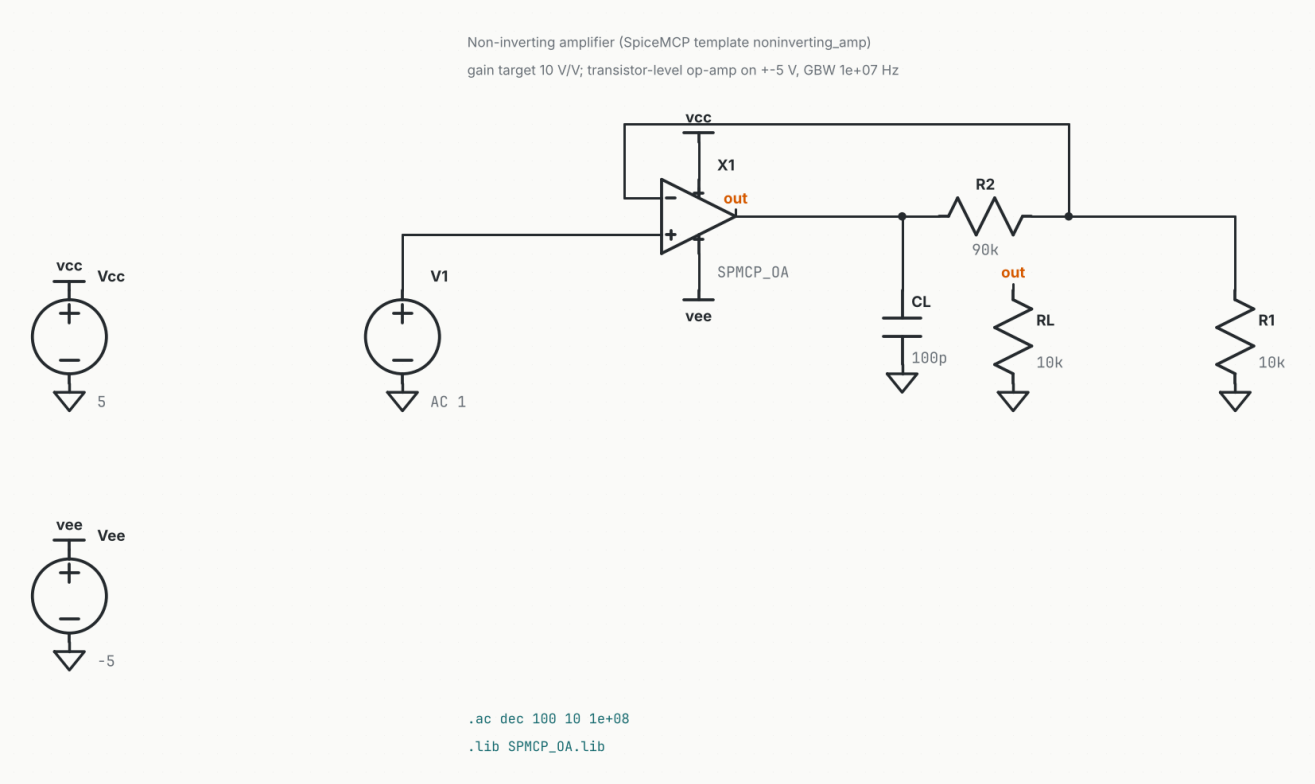
KEY RESULT

- -3 dB at 9998 Hz for a 10 kHz spec, flat passband (peak 0.001 dB)
- -40.3 dB one decade above cutoff (40 dB/decade)

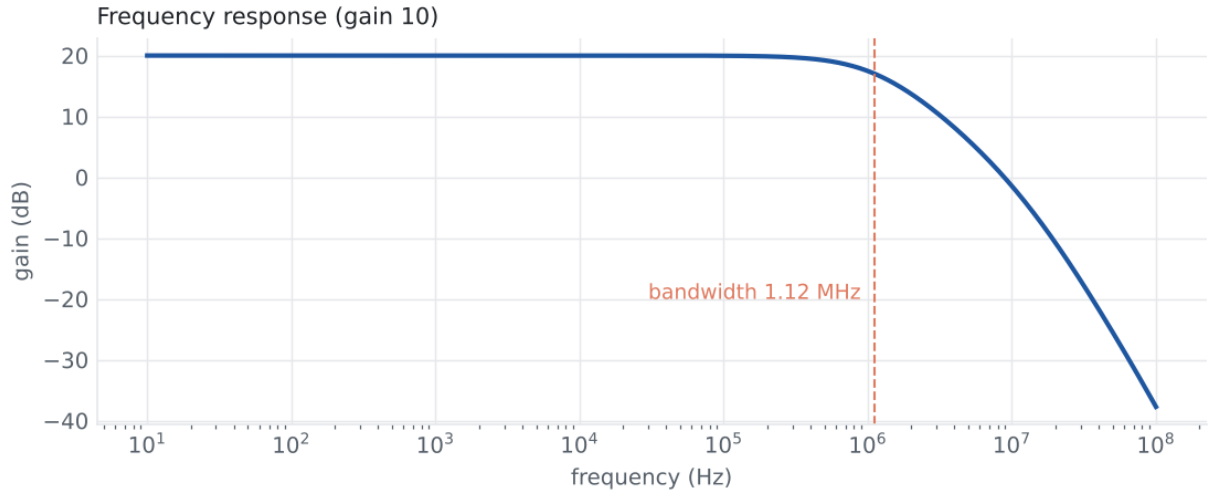
CIRCUIT 3 OF 25 · READY-MADE CIRCUIT

3. Non-inverting amplifier

Amplifies by 1 + R2/R1 (here x10) with a high input impedance.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

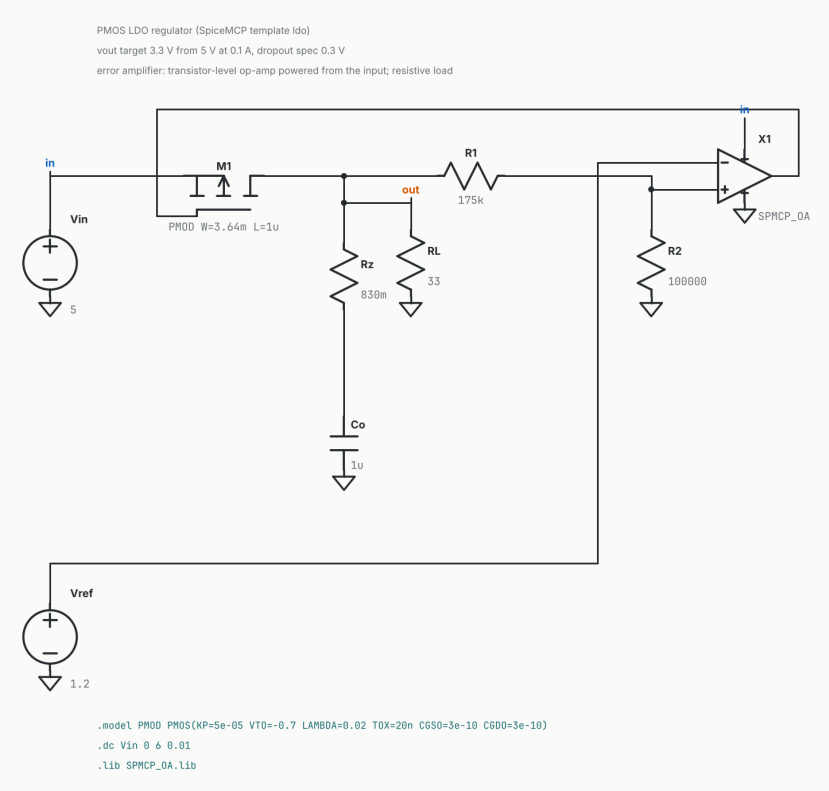
KEY RESULT

- Gain 20.00 dB (x10.00) for a x10 spec
- Bandwidth 1.12 MHz (spec: at least 100 kHz)
- Phase margin 85° (stable)

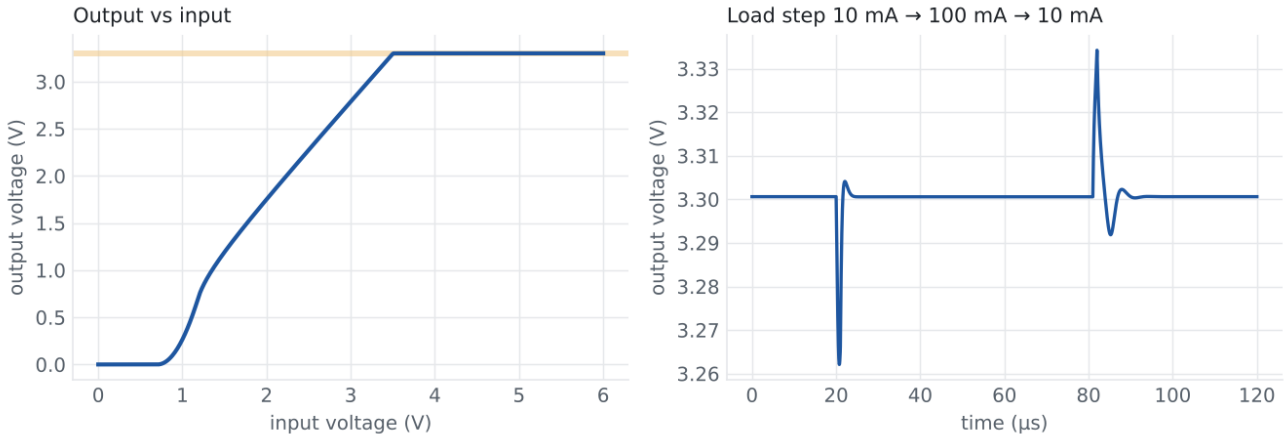
CIRCUIT 4 OF 25 · READY-MADE CIRCUIT

4. LDO voltage regulator

Linear regulator: 5 V in, 3.3 V out at 0.1 A, using a PMOS pass transistor and an error amplifier.



Schematic drawn by SpiceMCP · readability 76/100



Simulated response

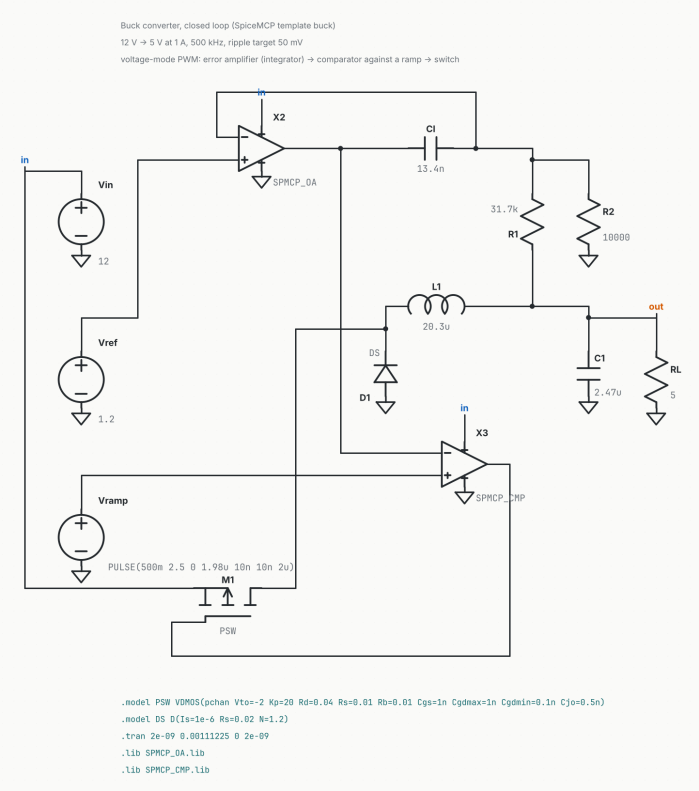
KEY RESULT

- 3.301 V out from 5 V in, at 0.1 A
- Still 3.300 V at 3.6 V in (low dropout)
- Phase margin 72°; load step dip 39 mV

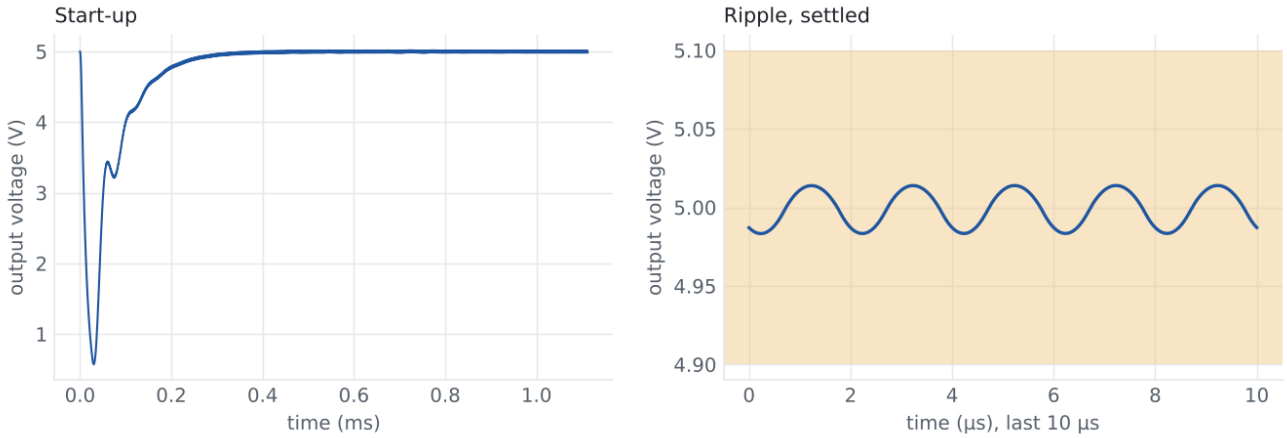
CIRCUIT 5 OF 25 · READY-MADE CIRCUIT

5. Buck DC-DC converter (closed loop)

Switching step-down converter: 12 V in, 5 V out at 1 A, with a PWM controller that holds the output.



Schematic drawn by SpiceMCP · readability 69/100



Simulated response

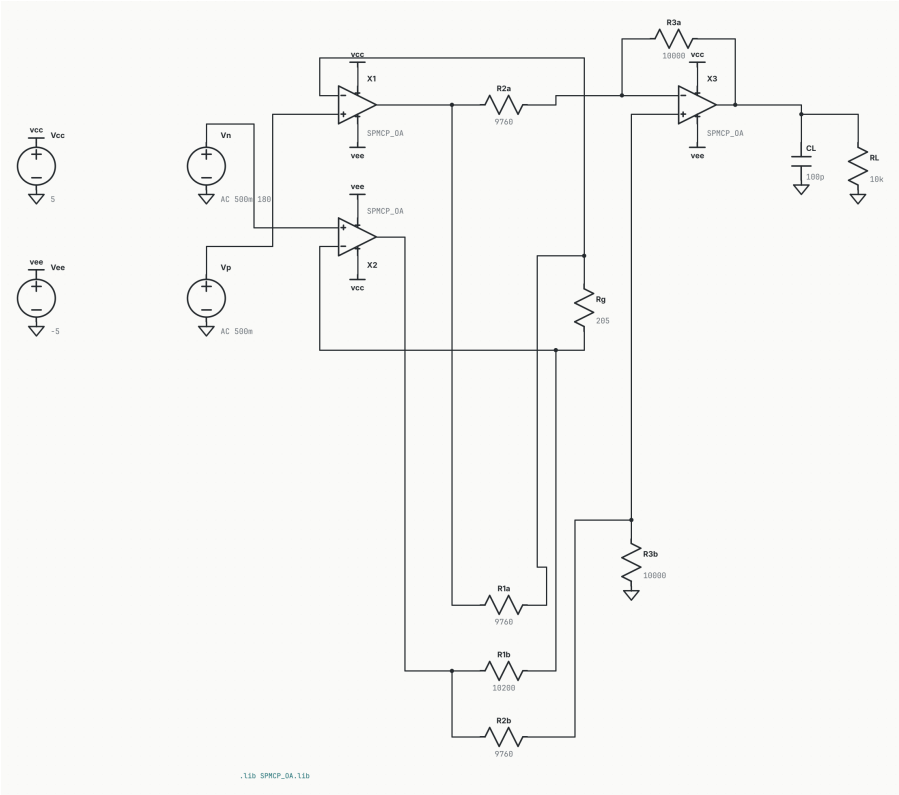
KEY RESULT

- 4.999 V out from 12 V in, at 1 A, 500 kHz switching
- Ripple 32.8 mV (limit 50 mV)
- Efficiency 89.5%

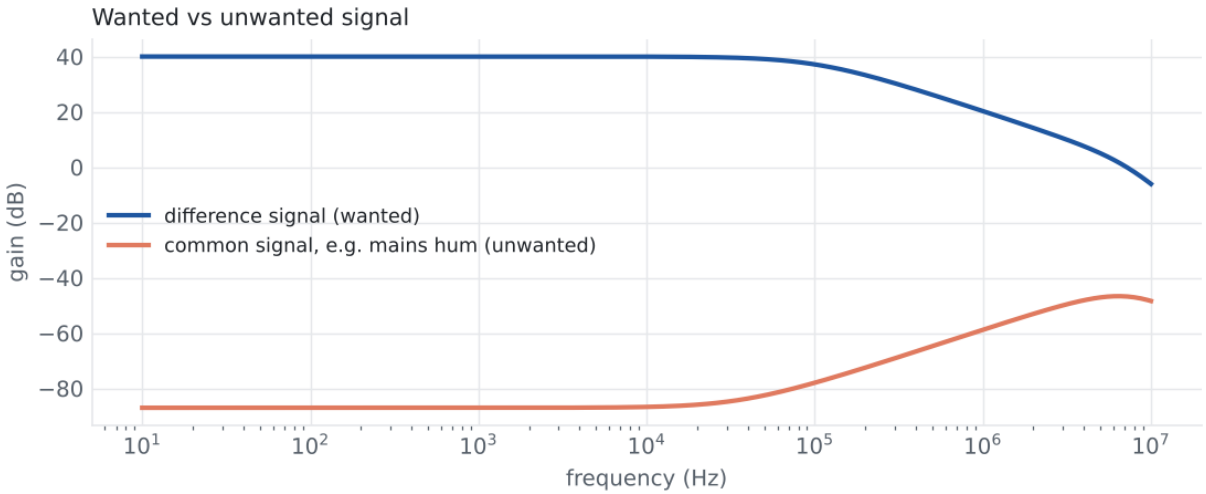
CIRCUIT 6 OF 25 · DESIGNED FROM SCRATCH

6. Instrumentation amplifier

Three op-amps: amplifies the difference between two sensor wires by 100 and rejects what is common to both (mains hum).



Schematic drawn by SpiceMCP · readability 26/100



Simulated response

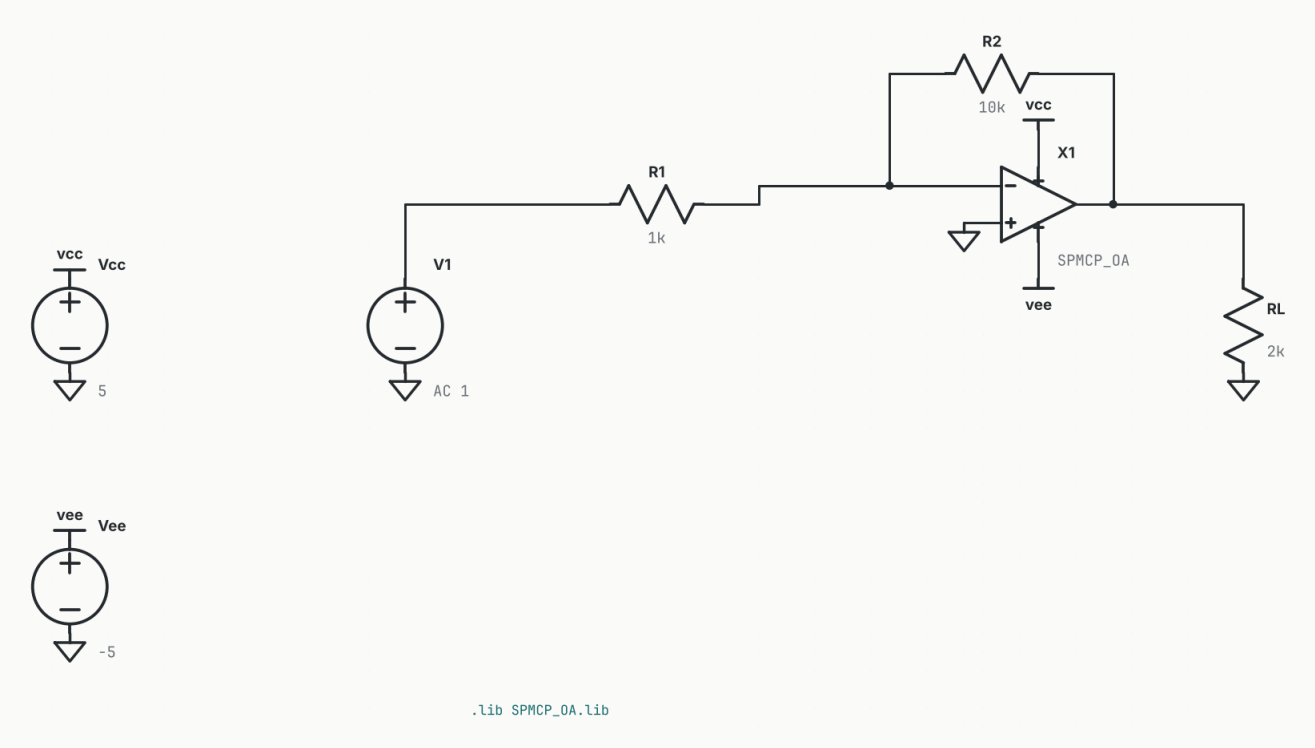
KEY RESULT

- Gain 40.00 dB (x100) for a x100 spec
- Bandwidth 104 kHz (spec: at least 20 kHz)
- Hum rejection (CMRR) at 60 Hz: 127 dB (spec: 80 dB)
- Finished with standard 1% values; 0.1% resistors give 100% of boards in spec

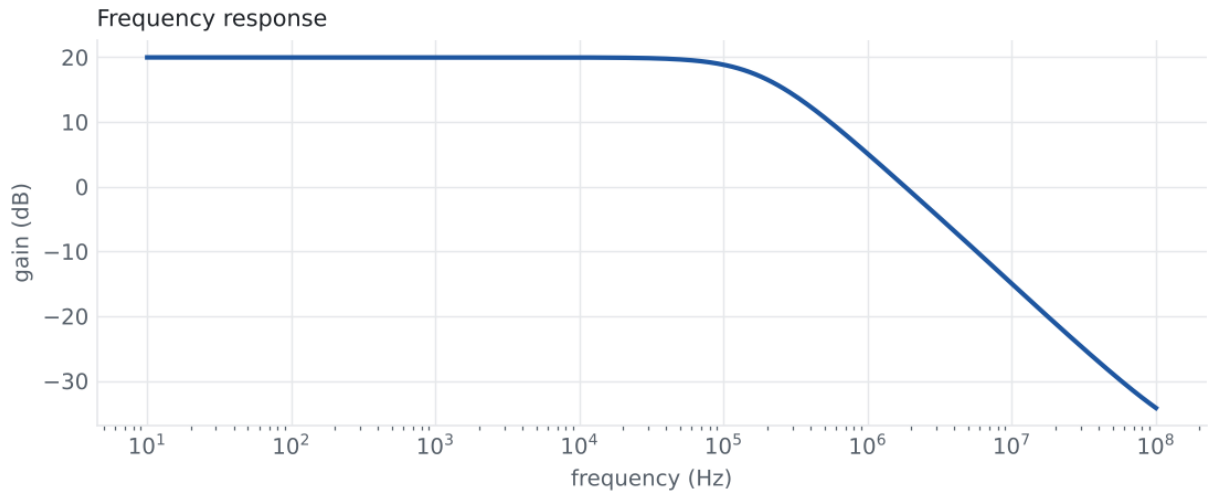
CIRCUIT 7 OF 25 · DRAWING TEST

7. Inverting amplifier

Amplifies by −R2/R1 (here −10).



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

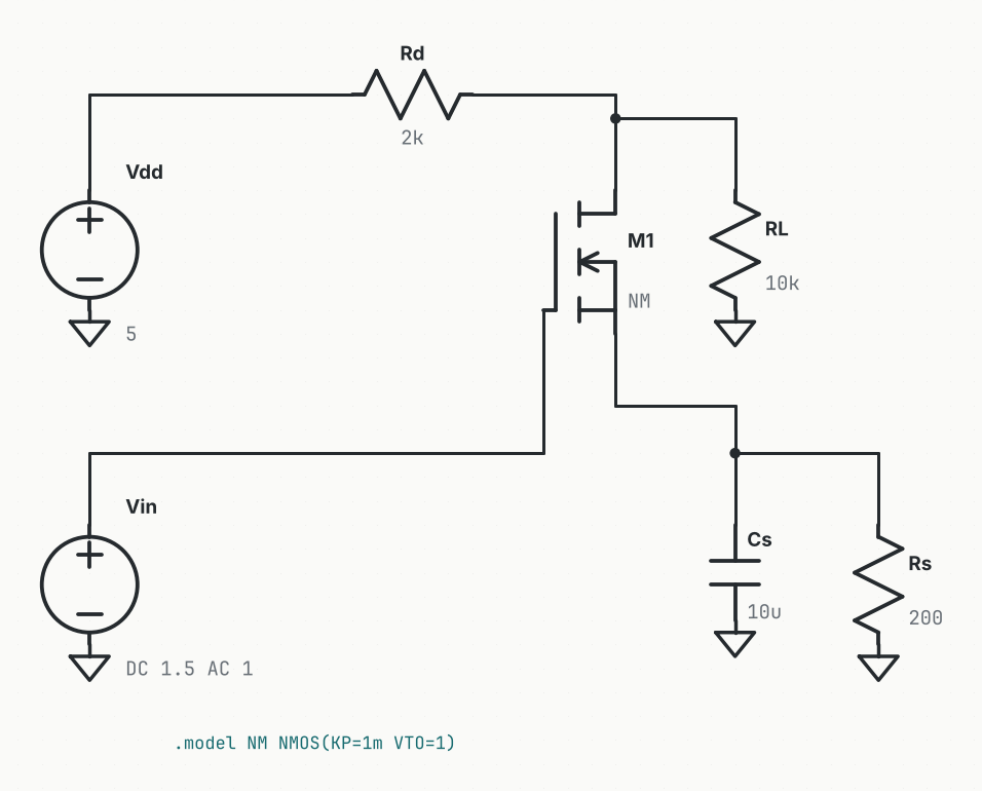
KEY RESULT

- Gain 19.98 dB (x−9.98), set by R2/R1 = 10
- Bandwidth 191 kHz (op-amp sized for 2 MHz gain-bandwidth)

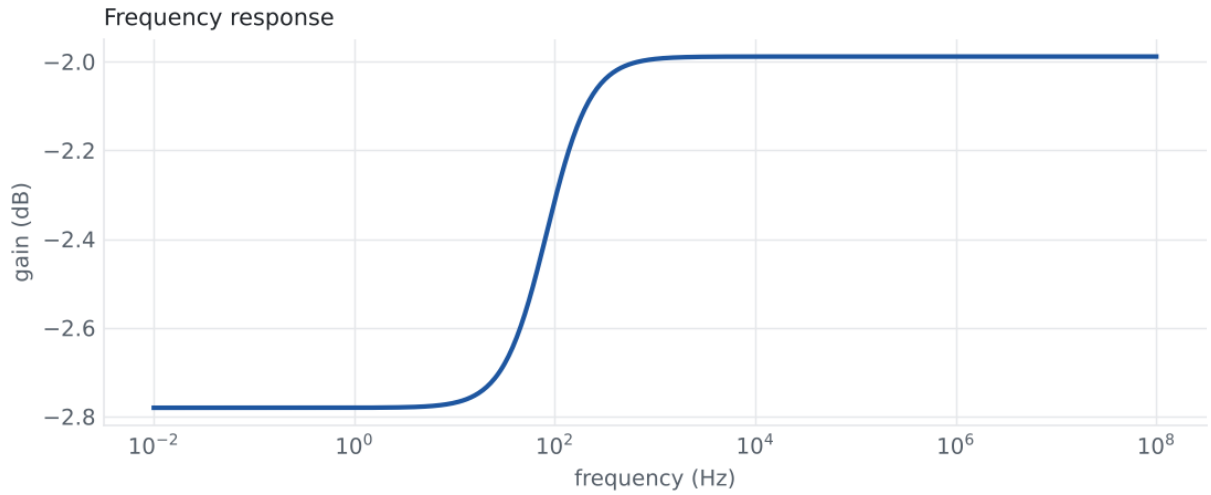
CIRCUIT 8 OF 25 · DRAWING TEST

8. Common-source MOSFET amplifier

Single-transistor voltage amplifier.



Schematic drawn by SpiceMCP · readability 99/100



Simulated response

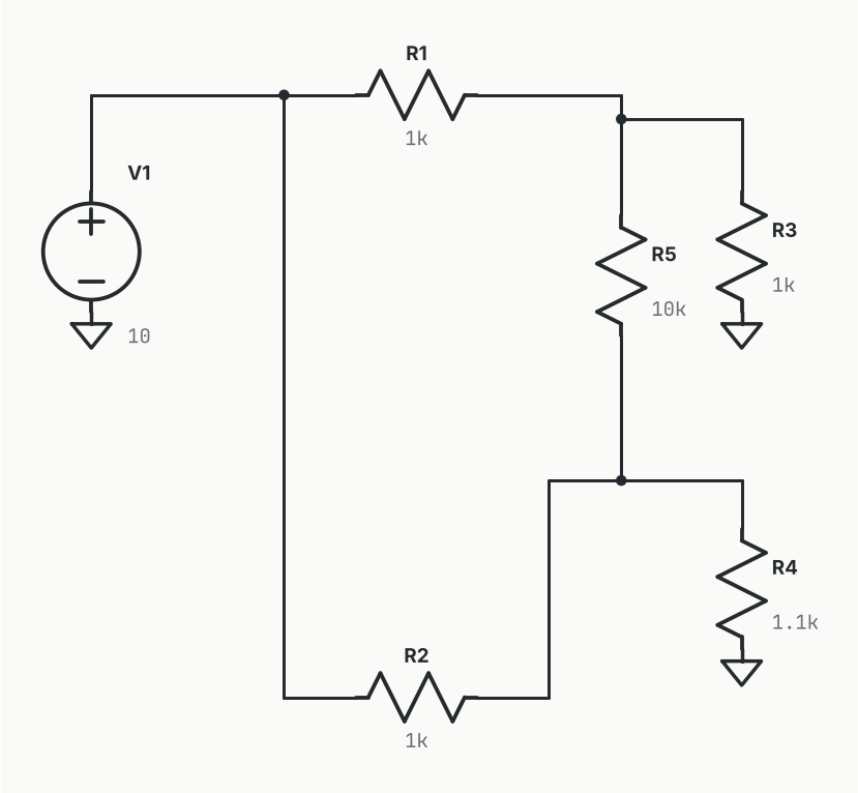
KEY RESULT

- Mid-band gain -2.0 dB (x0.80) from one MOSFET biased at 0.11 mA
- Below a few Hz the source resistor cuts the gain to -2.8 dB; the 10 µF bypass capacitor removes that loss above it

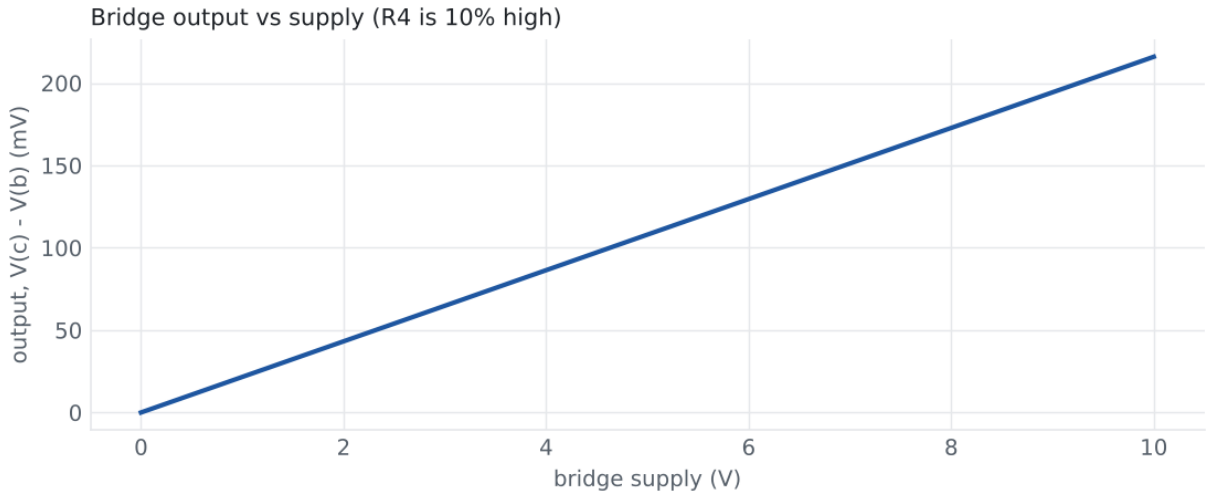
CIRCUIT 9 OF 25 · DRAWING TEST

9. Wheatstone bridge

Four-resistor bridge that turns a small resistance change (e.g. a strain gauge) into a voltage.



Schematic drawn by SpiceMCP · readability 99/100



Simulated response

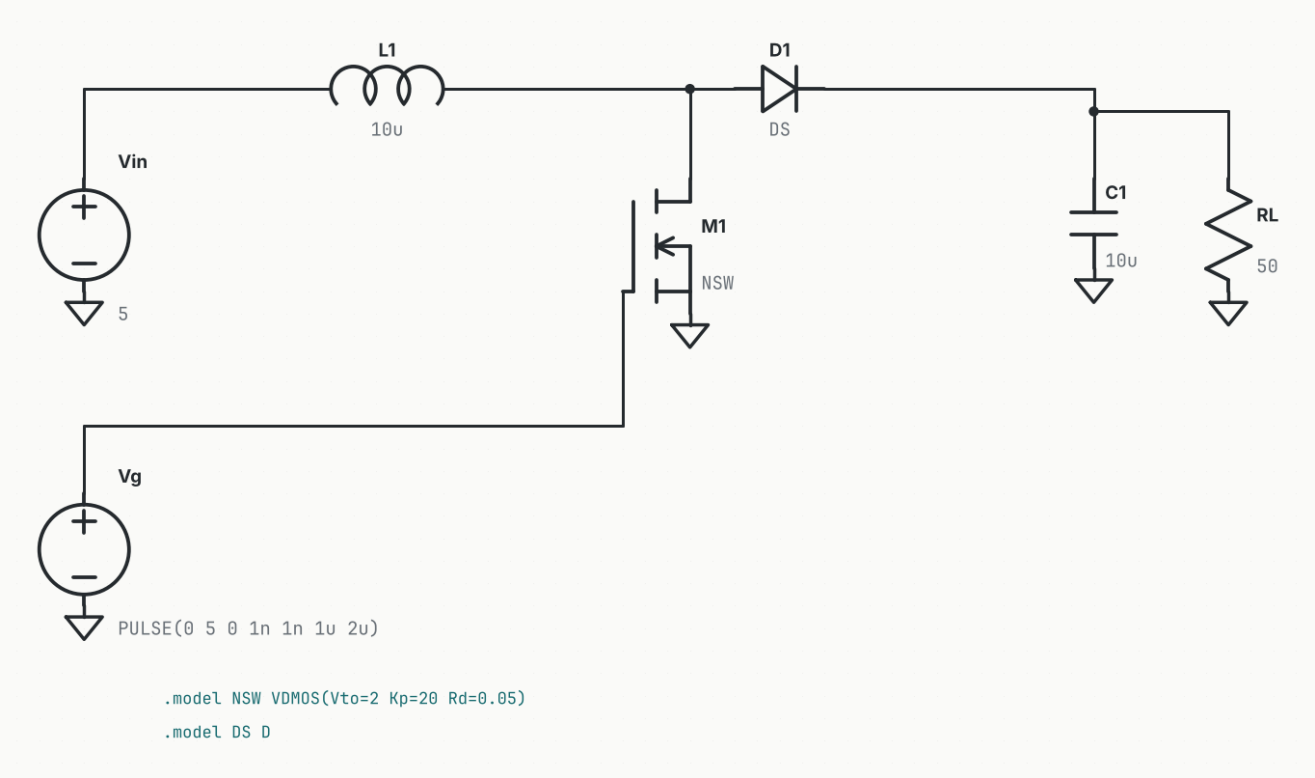
KEY RESULT

- A 10% change in one arm gives 216 mV across the bridge at 10 V
- Output is zero when all four arms match, and grows linearly with supply

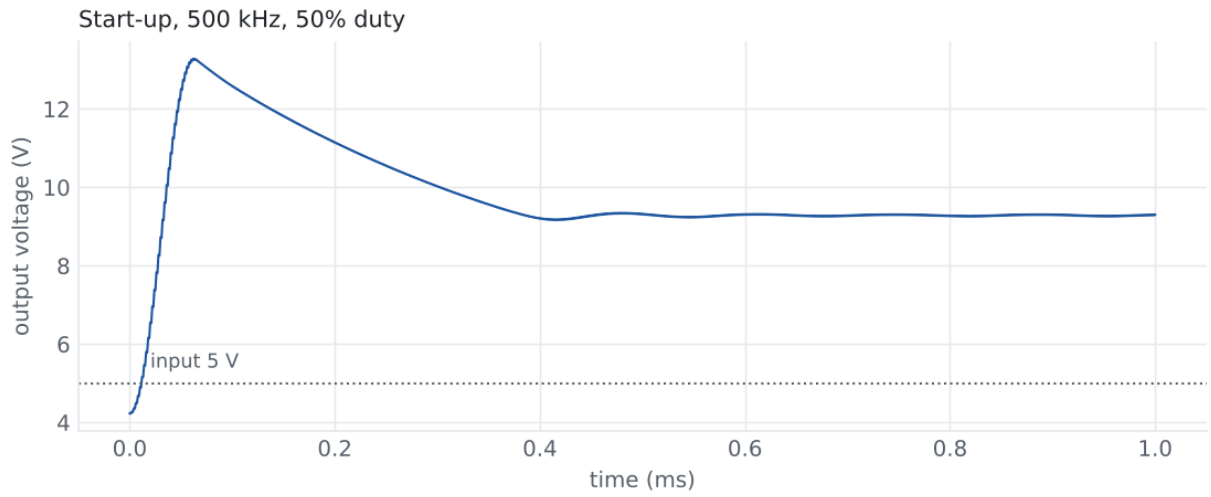
CIRCUIT 10 OF 25 · DRAWING TEST

10. Boost converter

Switching step-up converter: 5 V in, about 10 V out.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

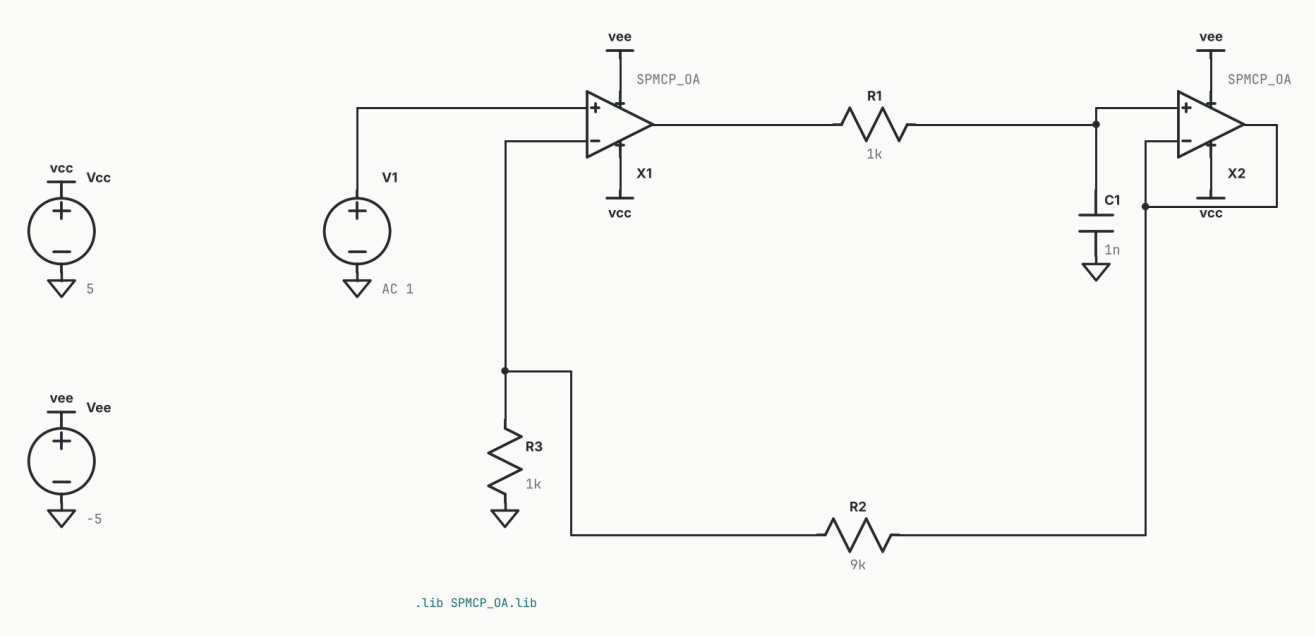
KEY RESULT

- Steps 5 V up to 9.28 V (ideal at 50% duty: 10 V) into 50 Ω
- Open loop: the output depends on the input; shows the drawer handles switchers

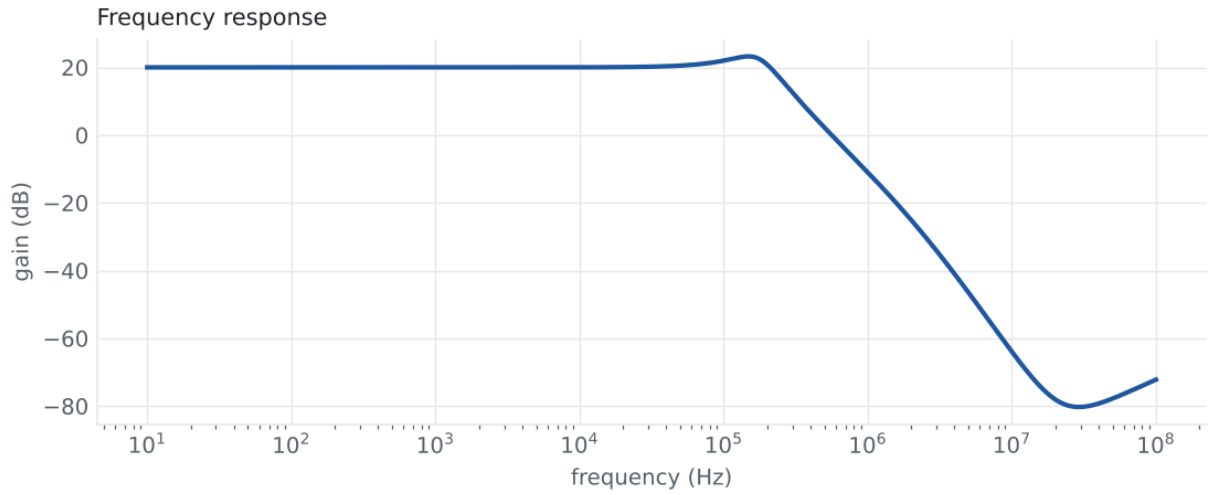
CIRCUIT 11 OF 25 · DRAWING TEST

11. Two-stage amplifier with feedback

Two op-amps with an RC filter between them, one feedback loop around both, gain 10.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

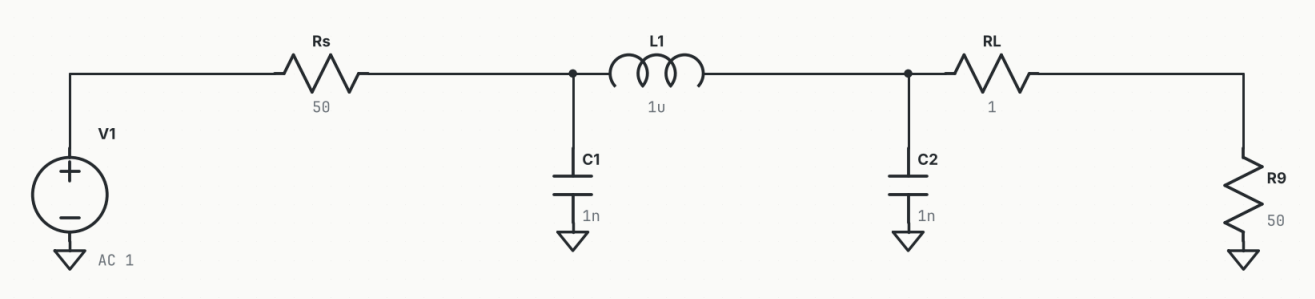
KEY RESULT

- Gain 20.00 dB (x10.00), set by 1 + R2/R3 = 10
- Feedback wraps both op-amps and the RC between them

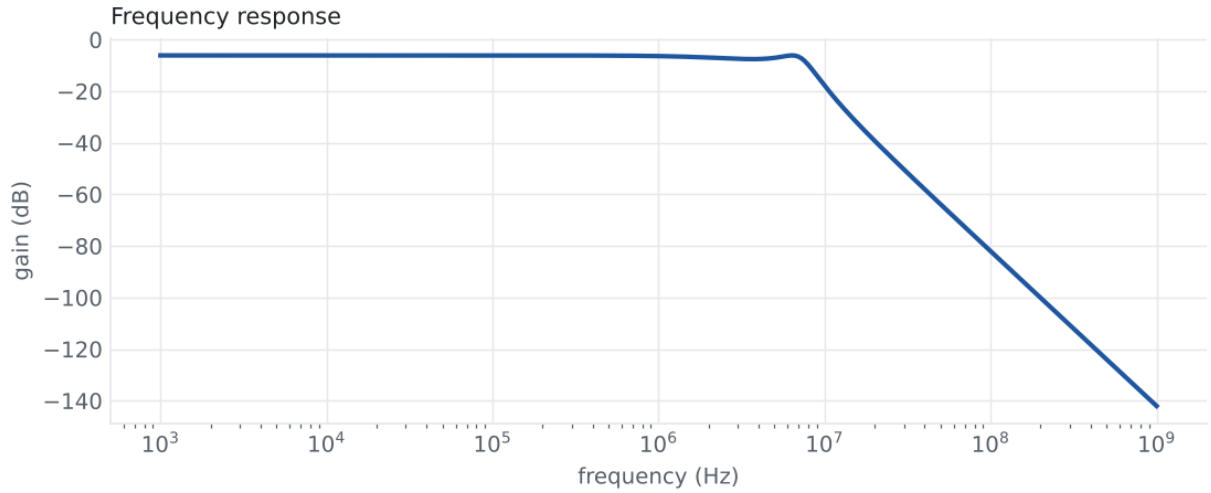
CIRCUIT 12 OF 25 · DRAWING TEST

12. Pi filter

C-L-C low-pass, as used on supply and RF lines.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

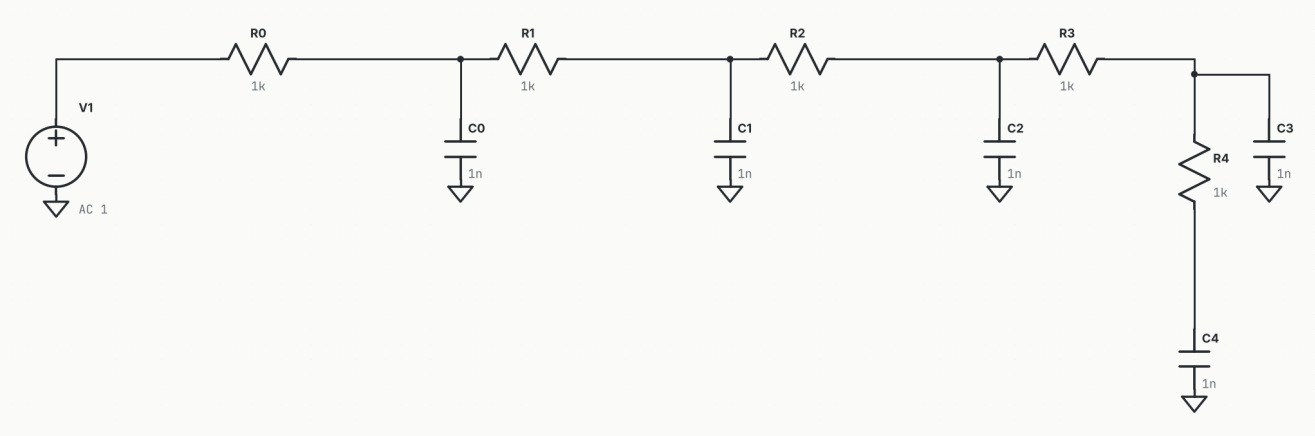
KEY RESULT

- Passband -6.1 dB (the 50 Ω source and load halve the voltage)
- -82 dB at 100 MHz: steep 3rd-order roll-off

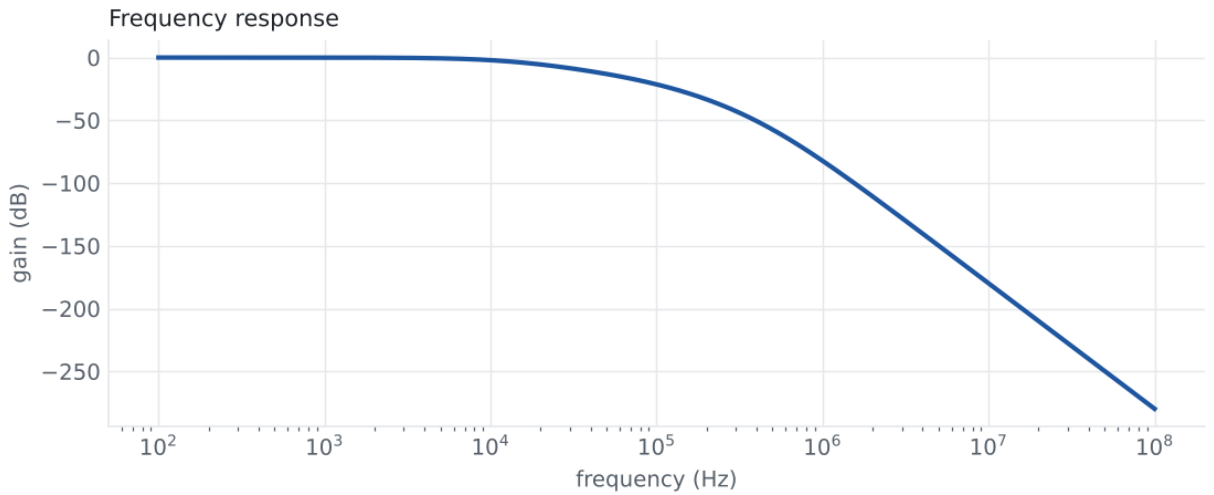
CIRCUIT 13 OF 25 · DRAWING TEST

13. 5-stage RC ladder

Five RC sections in a row: a steep passive low-pass.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

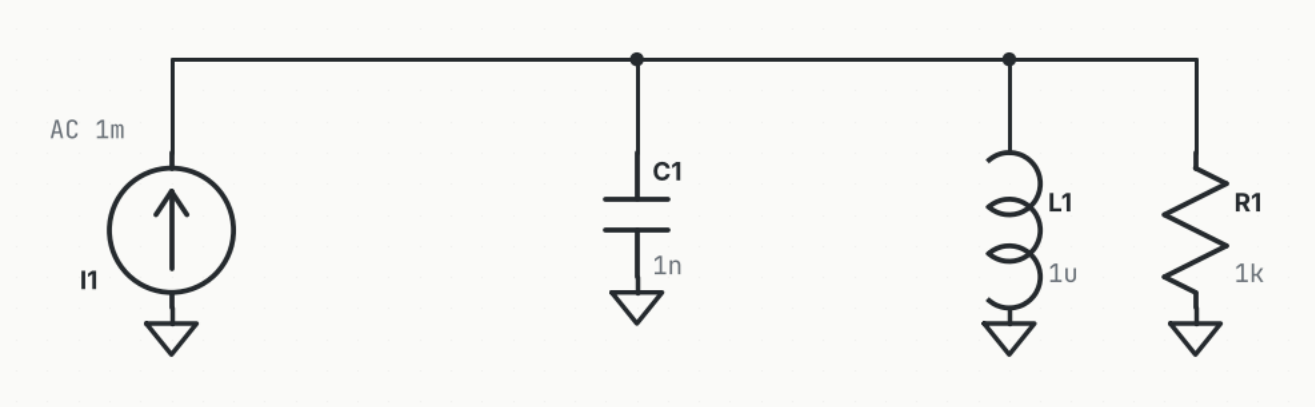
KEY RESULT

- -3 dB at 13.2 kHz (each section loads the one before it)
- -180 dB at 10 MHz: 100 dB/decade, five times one RC section

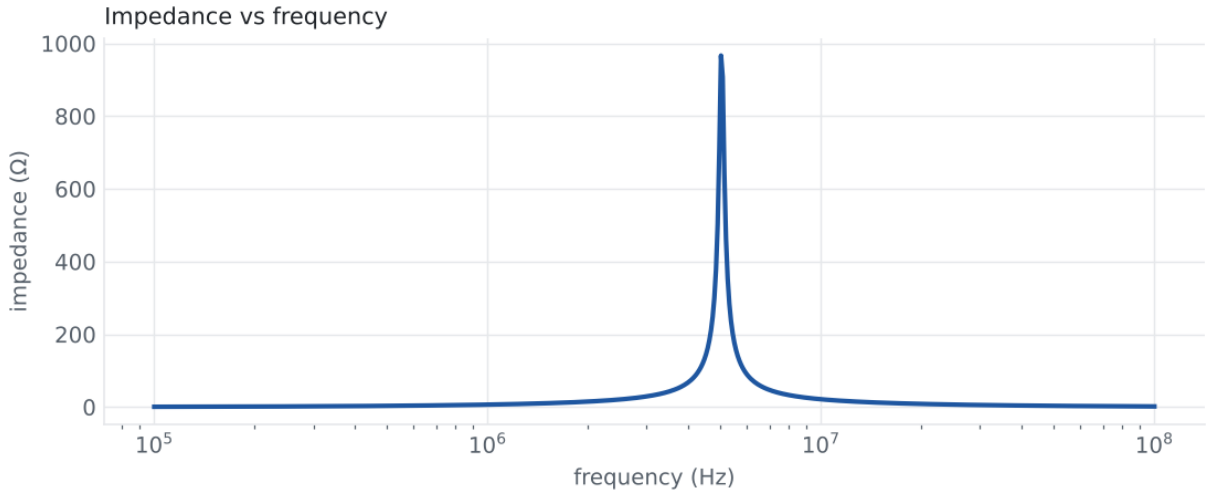
CIRCUIT 14 OF 25 · DRAWING TEST

14. LC tank

Inductor and capacitor in parallel: resonates at one frequency.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

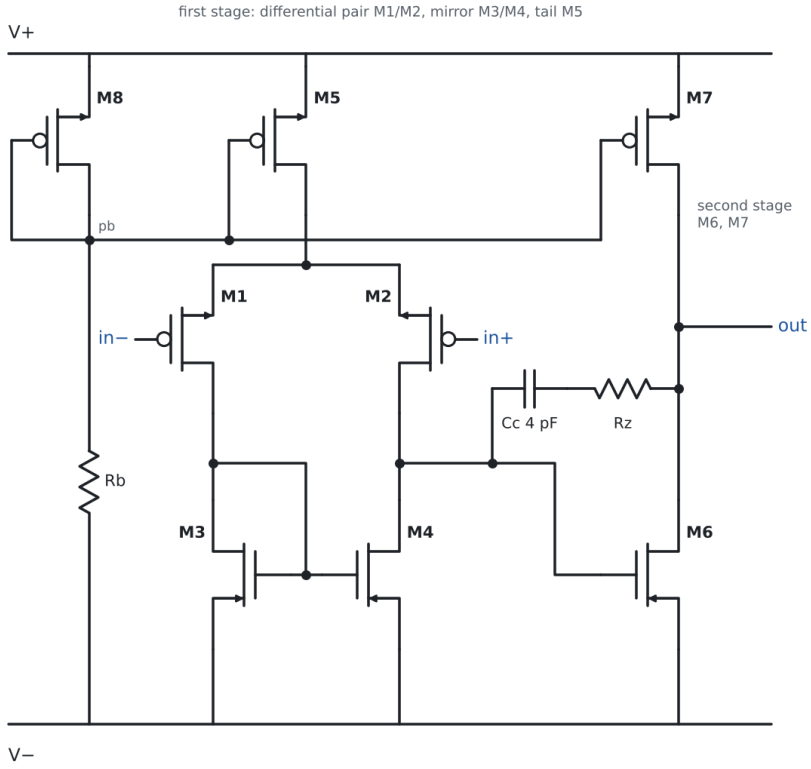
KEY RESULT

- Resonates at 5.01 MHz (1/2π√LC = 5.03 MHz)
- Peak impedance 967 Ω, set by the 1 kΩ loss resistor

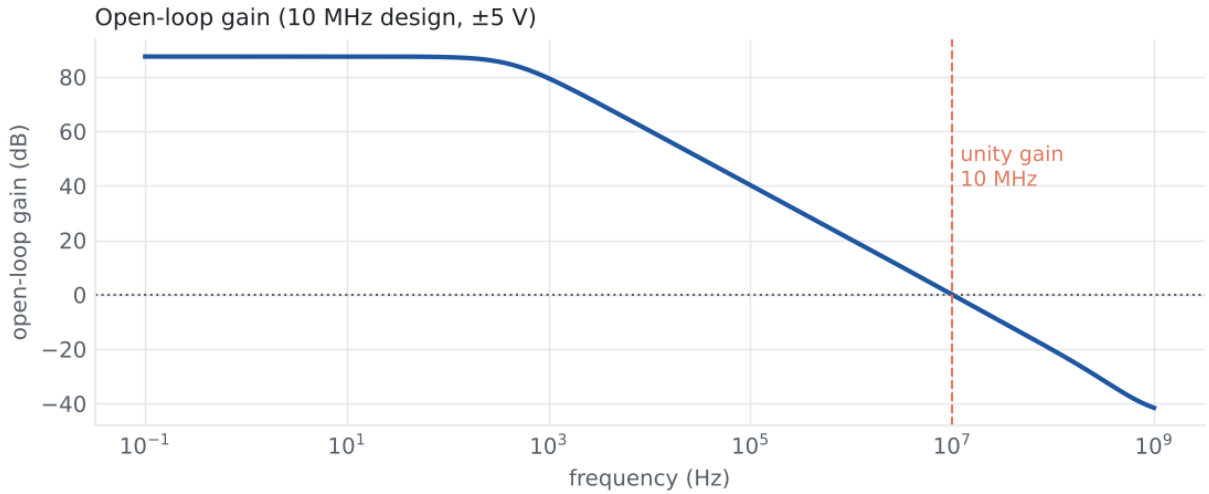
CIRCUIT 15 OF 25 · BUILDING BLOCK

15. Two-stage CMOS op-amp

The 8-MOSFET op-amp inside circuits 2, 3, 4, 6, 7 and 11: differential pair, current mirror, gain stage, Miller compensation.



Transistor-level schematic



Simulated response

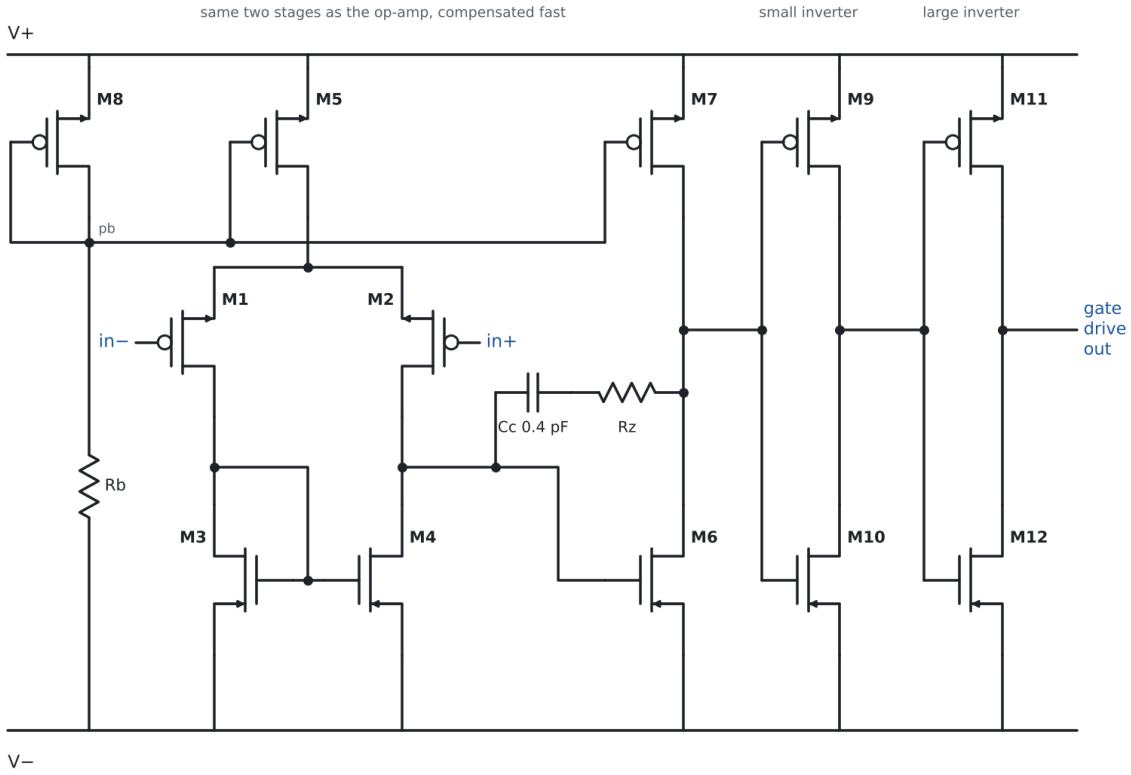
KEY RESULT

- Open-loop gain 87.5 dB
- Unity-gain frequency 10 MHz for a 10 MHz spec
- Sized from a target gain-bandwidth; used inside circuits 2, 3, 4, 5, 6, 7, 11

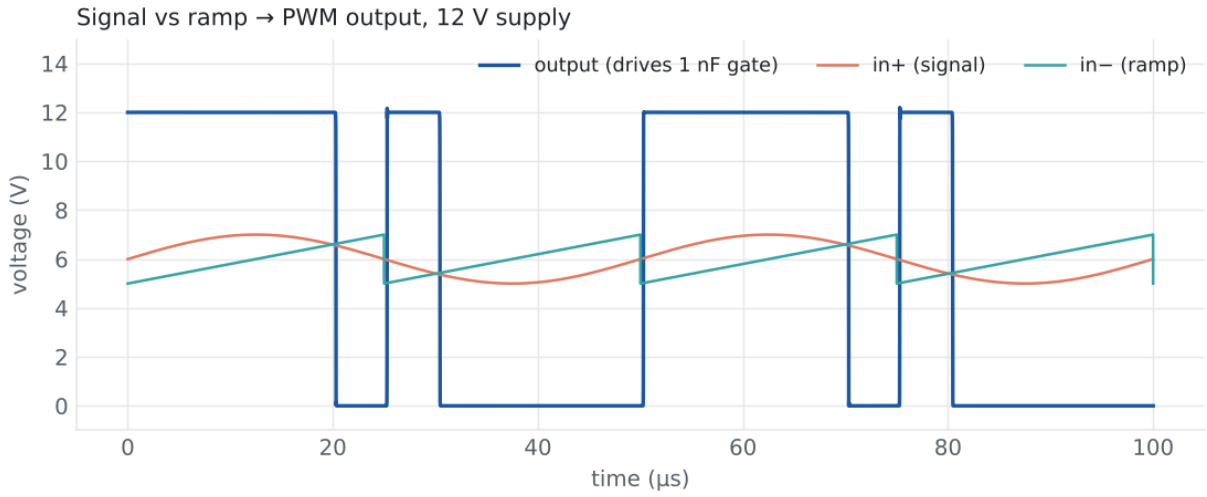
CIRCUIT 16 OF 25 · BUILDING BLOCK

16. Comparator with gate drive

The op-amp's two stages, fast, then two CMOS inverters (12 MOSFETs) that can switch a power MOSFET.



Transistor-level schematic



Simulated response

KEY RESULT

- Output swings rail to rail (0 to 12 V) whenever in+ crosses in−
- Tapered buffer (small then large inverter) drives a power MOSFET gate directly
- This is the PWM block of the closed-loop buck (circuit 5)

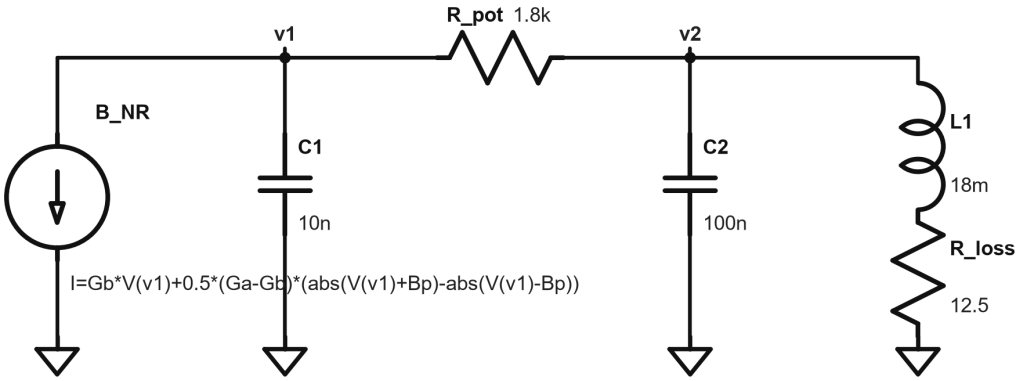
CIRCUIT 17 OF 25 · DEMO

17. Chua's chaotic circuit

The simplest electronic circuit that behaves chaotically.

Chua's Circuit - Canonical Chaotic Resonator

Double-Scroll Oscillator: C1=10nF, C2=100nF, L1=18mH (rL=12.5 ohm), R_pot=1.8k

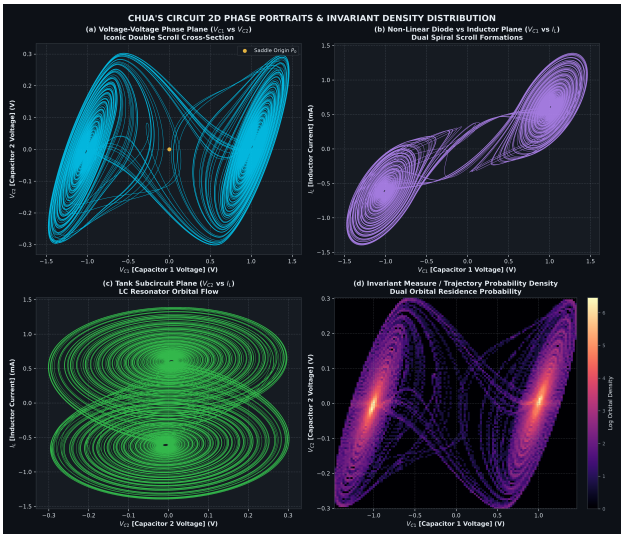


.param Ga=-0.75757m Gb=-0.40909m Bp=1.0

.tran 0 60m 10m 2u uic

.options plotwinsize=0

Schematic (gallery)



Simulated phase portraits: the double scroll

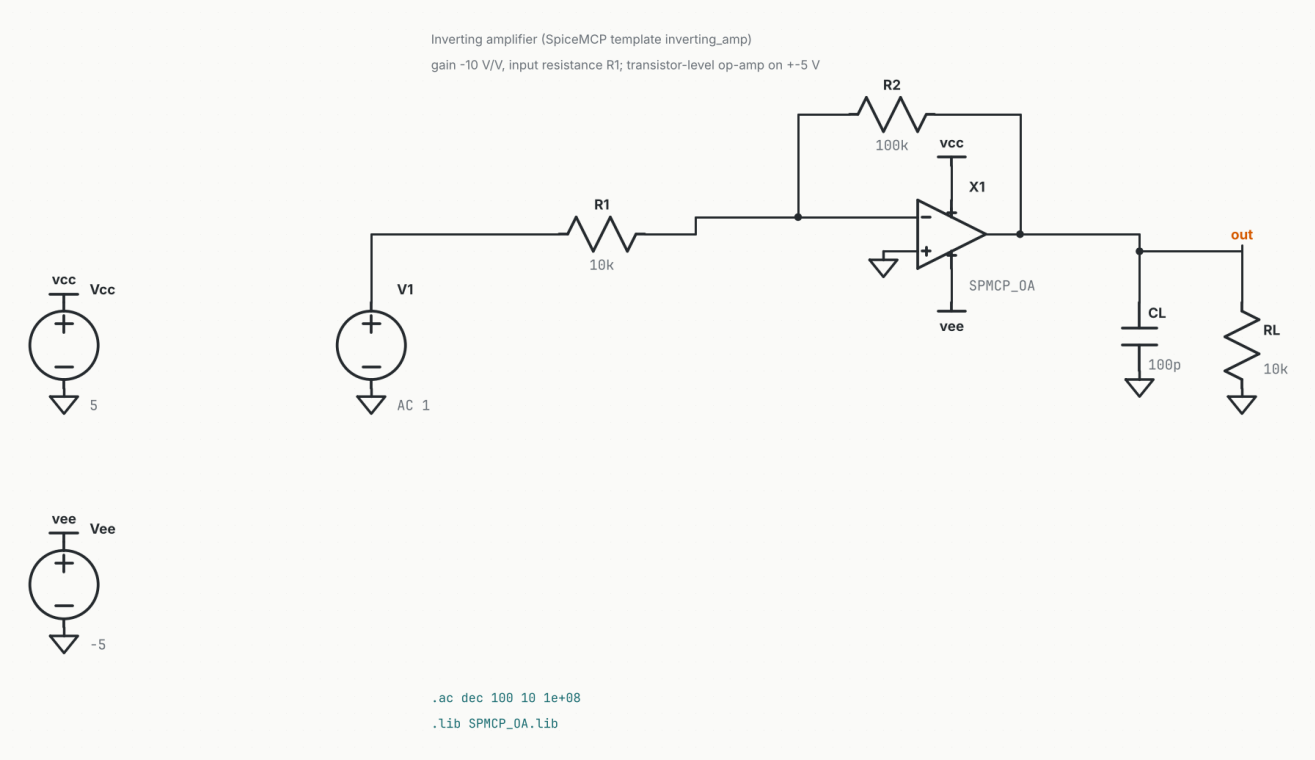
KEY RESULT

- Chaotic oscillator: L, two capacitors and a negative-resistance (Chua) diode
- Simulated in LTspice; the output never repeats and traces a double scroll
- More plots (bifurcation, butterfly effect) in the README gallery

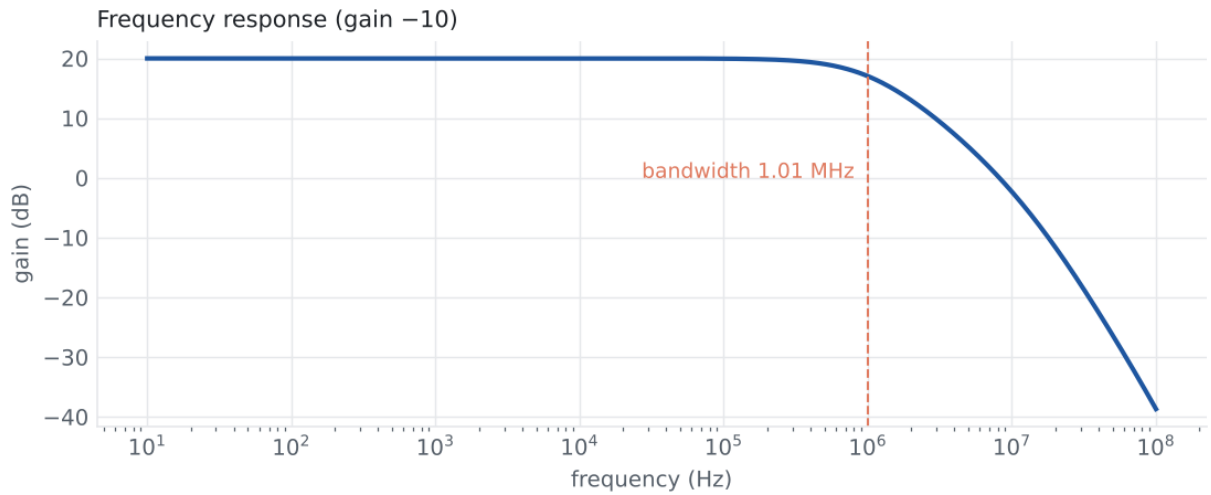
CIRCUIT 18 OF 25 · READY-MADE CIRCUIT

18. Inverting amplifier (template)

Amplifies by −R2/R1 with a set input resistance; sized from gain, bandwidth and load.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

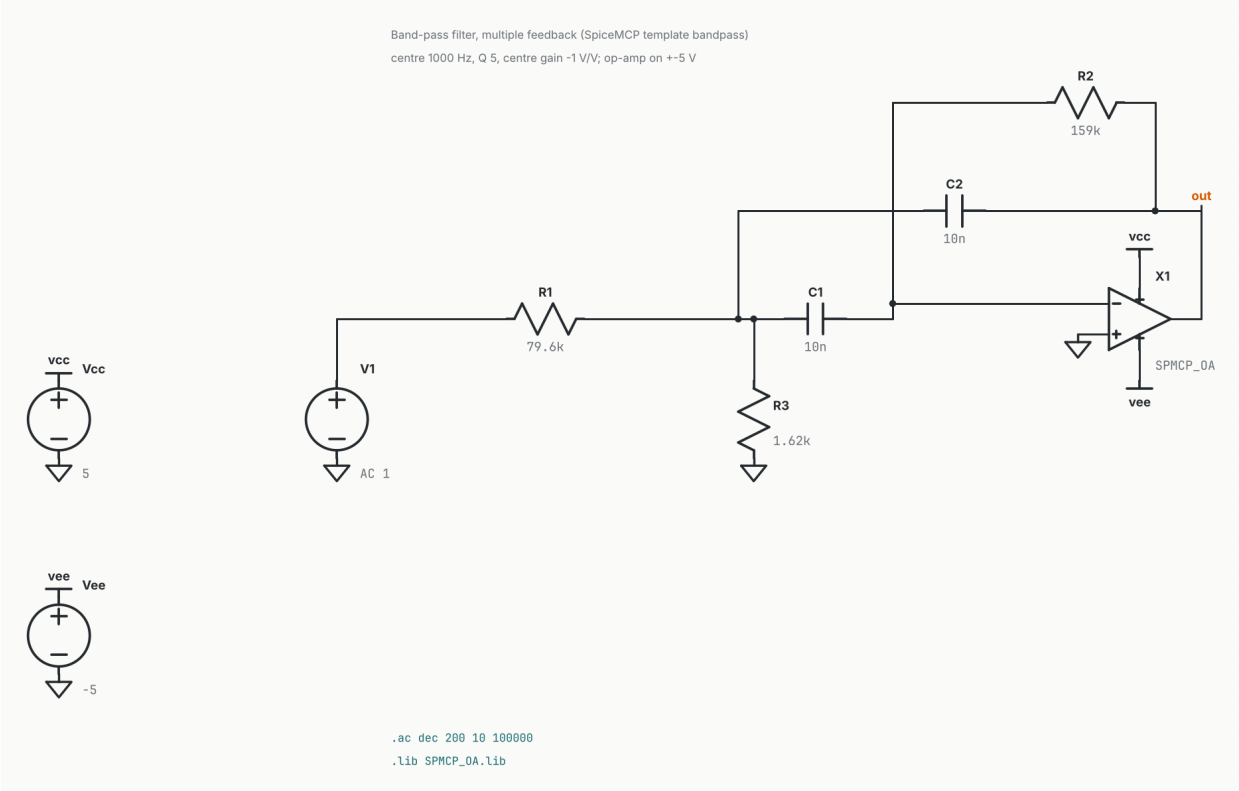
KEY RESULT

- Gain 20.00 dB (x−9.995) for a x−10 spec, input resistance 10 kΩ
- Bandwidth 1.01 MHz (spec: at least 100 kHz); phase margin 86°

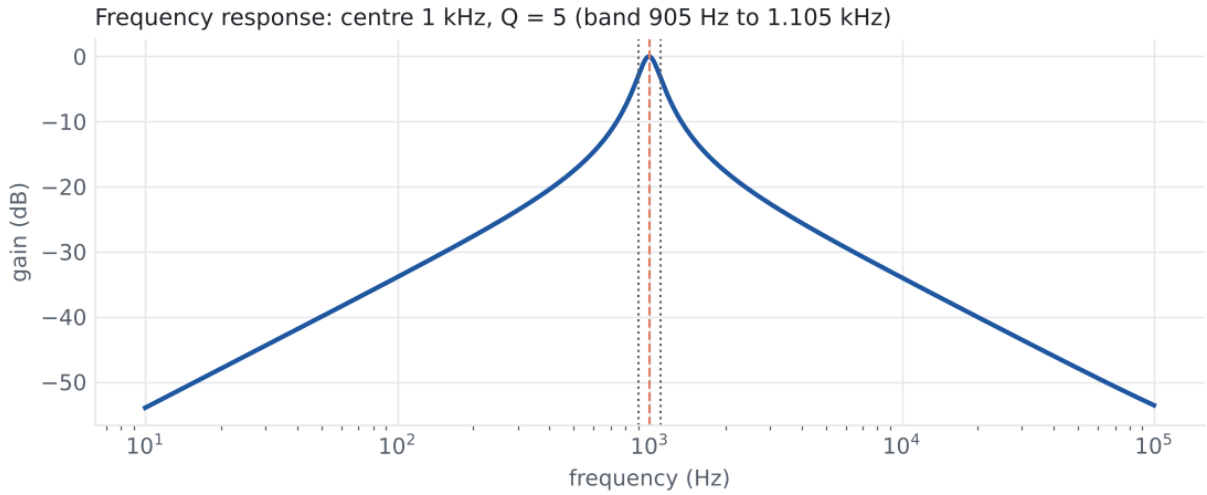
CIRCUIT 19 OF 25 · READY-MADE CIRCUIT

19. Band-pass filter

Multiple-feedback band-pass: passes a band around one centre frequency (here 1 kHz, Q = 5).



Schematic drawn by SpiceMCP · readability 88/100



Simulated response

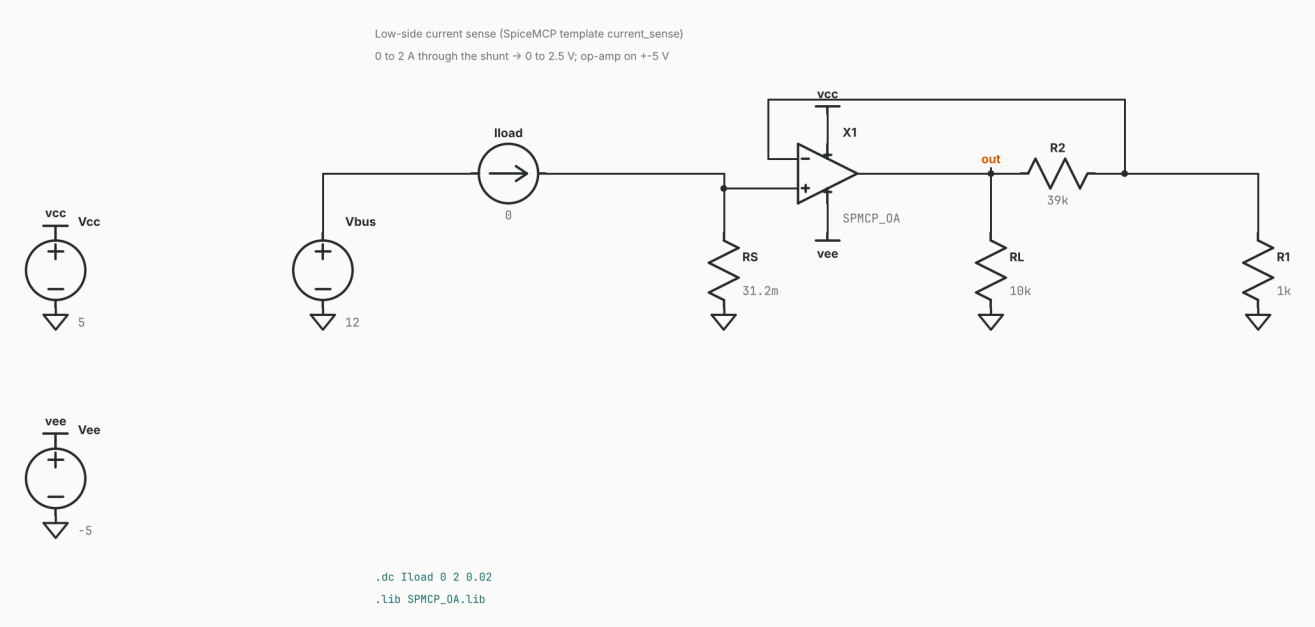
KEY RESULT

- Centre gain -0.01 dB at 1 kHz for a 0 dB spec (peak -0.01 dB)
- Response at the band edges -2.81 and -3.23 dB (−3 dB ideal, Q = 5); phase margin 90°

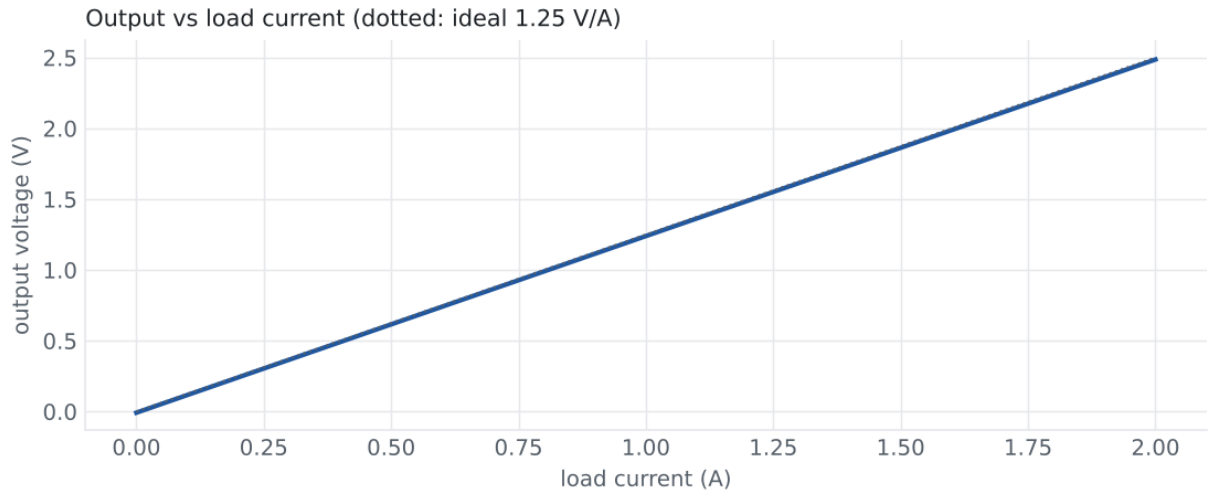
CIRCUIT 20 OF 25 · READY-MADE CIRCUIT

20. Current-sense amplifier

Low-side shunt and difference amplifier: turns a load current (here 0 to 2 A) into a voltage for an ADC.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

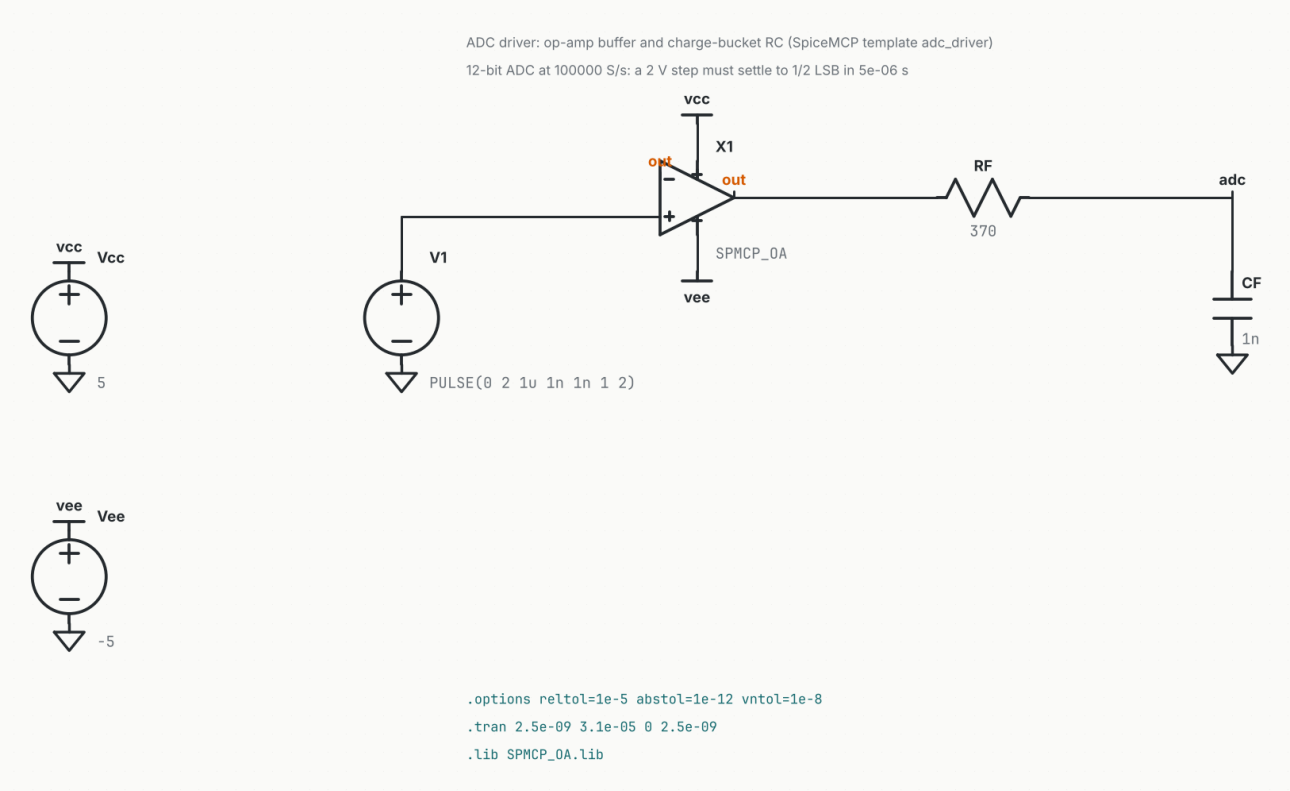
KEY RESULT

- 2.487 V out at 2 A for a 2.5 V full scale (error -0.53%)
- Error at 10% of full scale -3.39% (op-amp offset); bandwidth 104 kHz

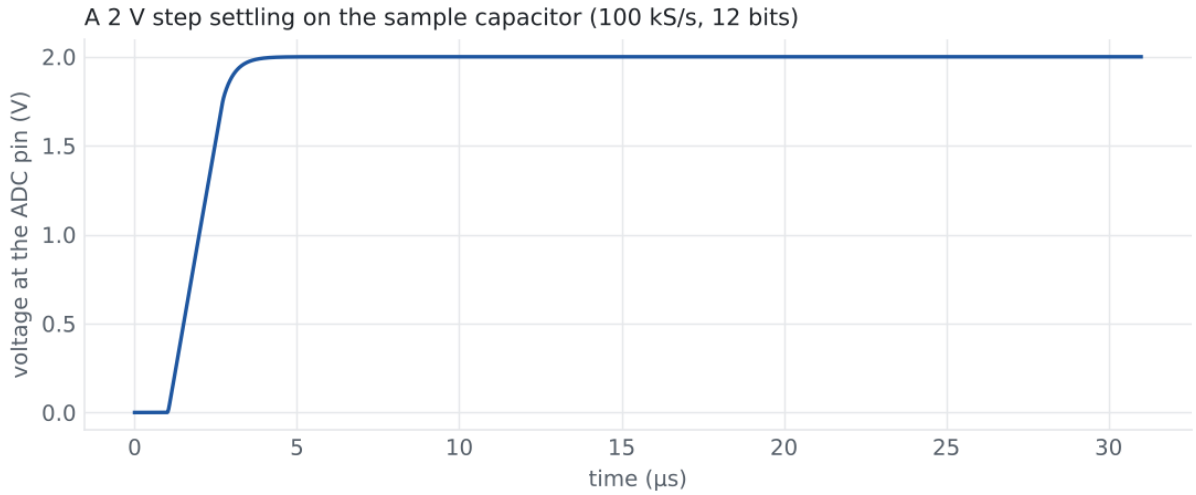
CIRCUIT 21 OF 25 · READY-MADE CIRCUIT

21. ADC driver

Op-amp buffer and RC kick-back filter that settles a sampling ADC's input to half an LSB in time.



Schematic drawn by SpiceMCP · readability 100/100



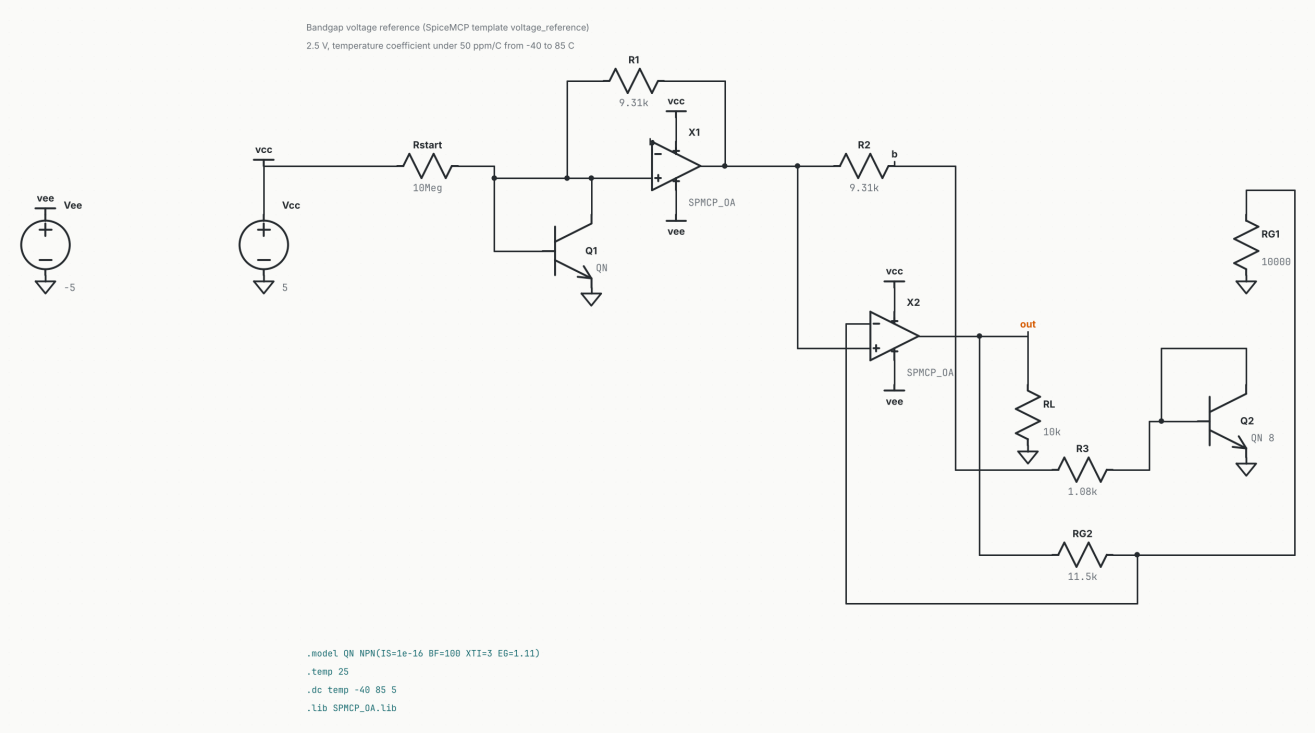
Simulated response

KEY RESULT

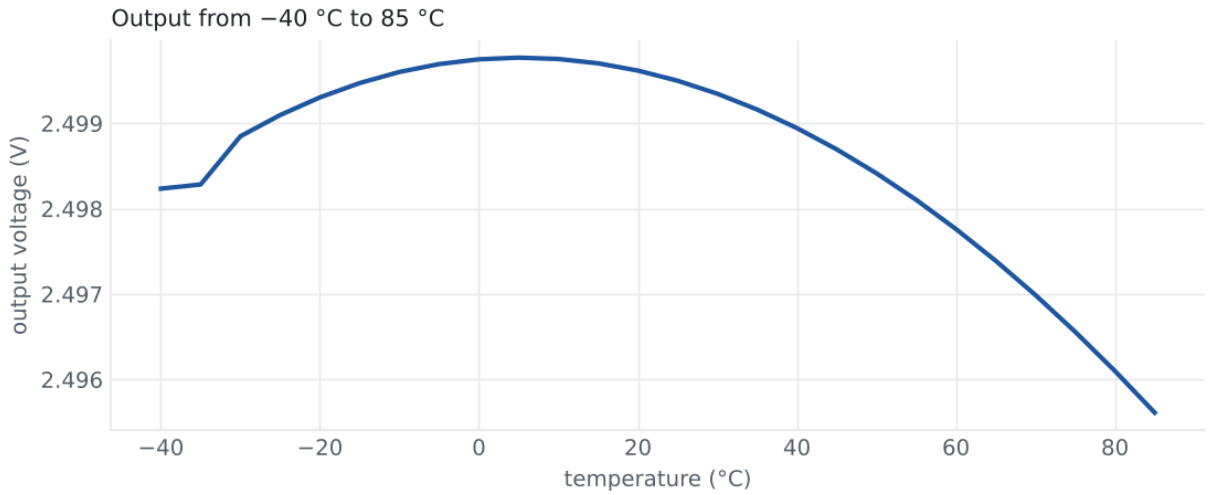
- Settles to 0.06 LSB of its final value within the acquisition time after a 2 V step (spec: half an LSB)
- Overshoot 0.00 LSB; fixed offset -0.40 LSB (the op-amp's); phase margin 87°

22. Bandgap voltage reference

Two transistors whose temperature drifts cancel: a steady voltage (here 2.5 V) from −40 to 85 °C.



Schematic drawn by SpiceMCP · readability 59/100



Simulated response

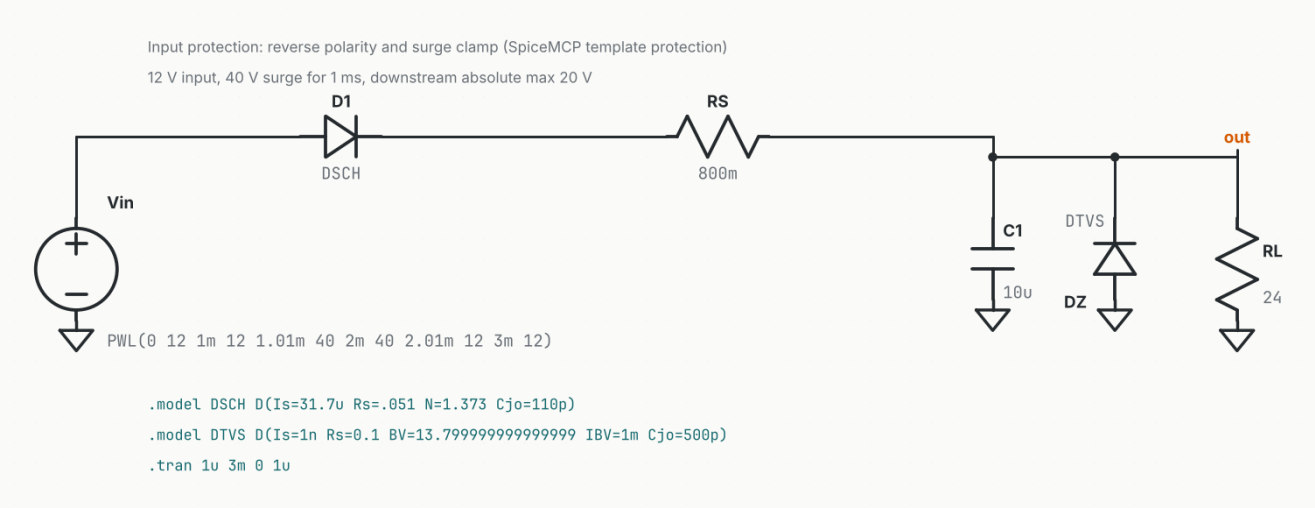
KEY RESULT

- 2.4995 V at 25 °C for a 2.5 V spec
- Drift 13.3 ppm/°C from −40 to 85 °C (spec: 50 ppm/°C)

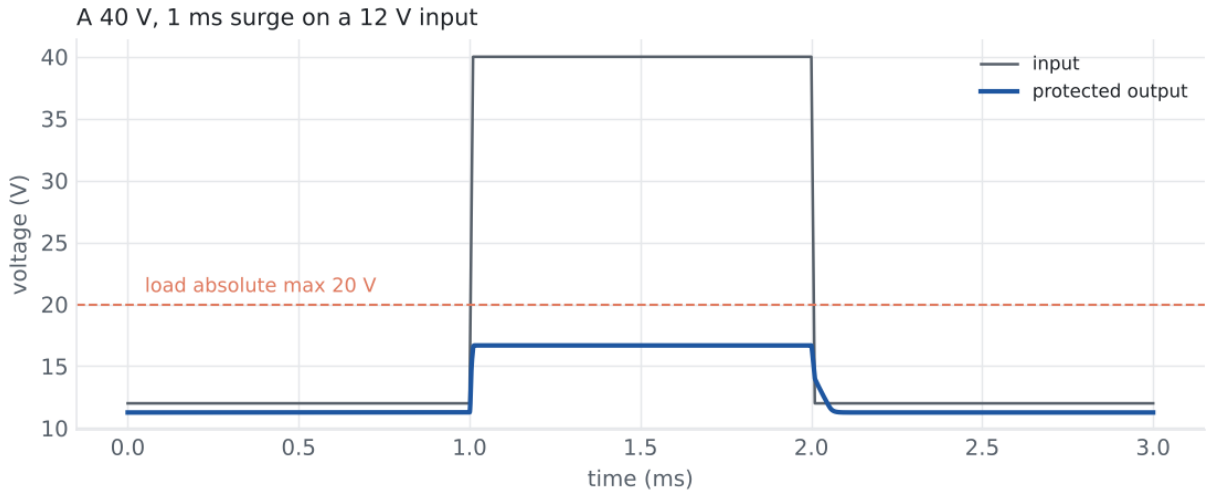
CIRCUIT 23 OF 25 · READY-MADE CIRCUIT

23. Input protection

Schottky diode against a reversed supply plus a TVS clamp against surges, in front of a 12 V load.



Schematic drawn by SpiceMCP · readability 100/100



Simulated response

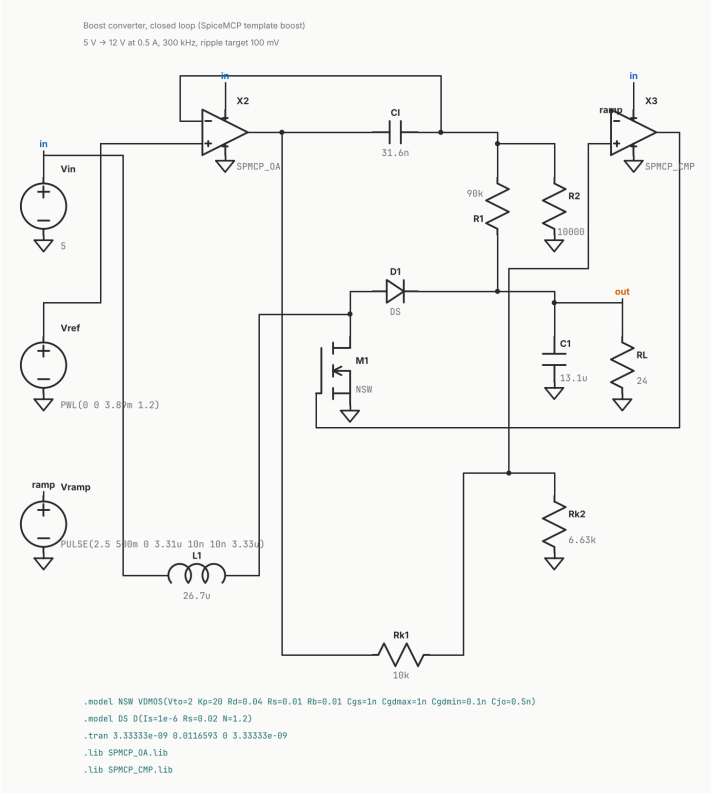
KEY RESULT

- Clamps a 40 V surge to 16.7 V (load max 20 V)
- Drops 0.74 V at 0.5 A; reversed input gives -0.001 V at the load

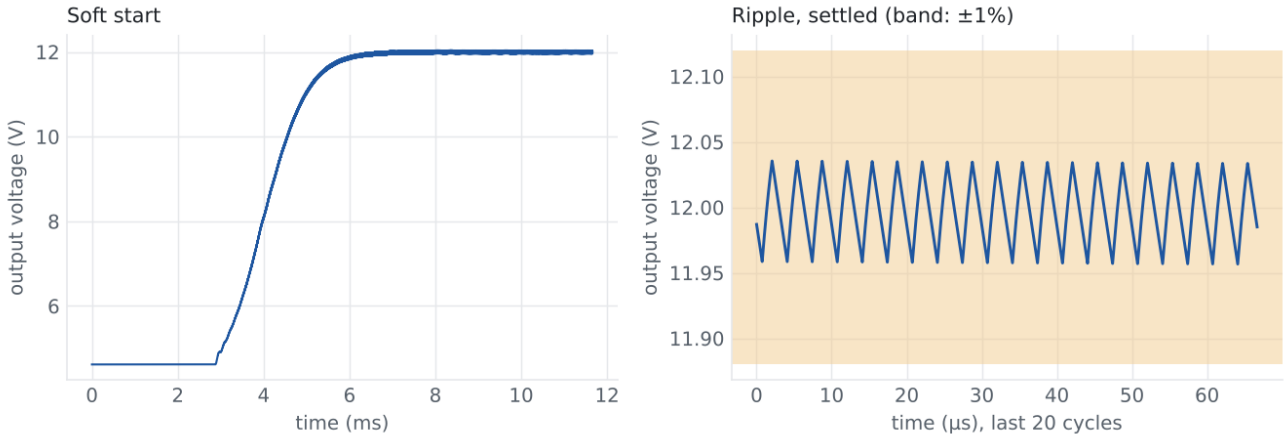
CIRCUIT 24 OF 25 · READY-MADE CIRCUIT

24. Boost converter (closed loop)

Switching step-up converter: 5 V in, 12 V out at 0.5 A, with soft start and a duty limit.



Schematic drawn by SpiceMCP · readability 43/100



Simulated response

KEY RESULT

- 11.997 V out from 5 V in, at 0.5 A, 300 kHz, closed loop
- Ripple 80 mV (limit 100 mV); efficiency 94%; phase margin 88°

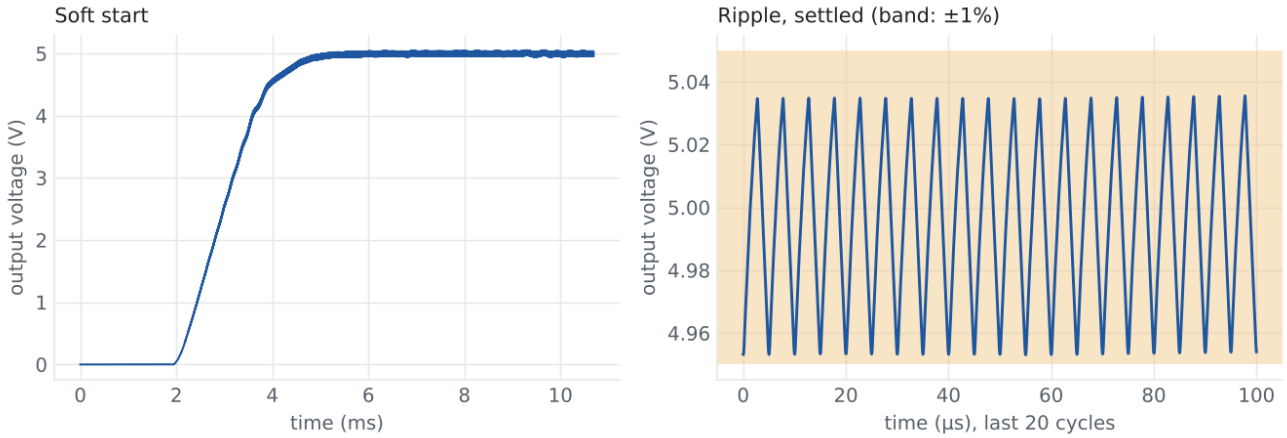
25. Flyback converter (closed loop)

Transformer-based converter: 12 V in, 5 V out at 1 A, with an RCD clamp on the switch.

Not drawn: the auto-placer has no symbol for the coupled windings (K1).
Power stage from the netlist:

```
Vin in 0 12
Vref ref 0 PWL(0 0 0.00356339 1.2)
Vramp ramp 0 PULSE(2.5 0.5 0 4.98e-06 10n 10n 5e-06)
X2 ref fb in 0 vc SPMCP_OA
CI vc fb {CI}
Rk1 vc vcd 10k
Rk2 vcd 0 1650.49
X3 vcd ramp in 0 g SPMCP_CMP
R1 out fb {R1}
R2 fb 0 10000
LP in sw {LP}
LS 0 sx {{LP}*0.472656}
K1 LP LS 0.98
M1 sw g 0 NSW
```

Not drawn by the auto-placer



Simulated response

KEY RESULT

- 4.997 V out from 12 V in, at 1 A, 200 kHz, closed loop
- Ripple 95 mV (limit 100 mV); switch sees 29.4 V with the RCD clamp (limit 60 V); phase margin 87°